Probing cosmic structure growth with galaxy peculiar velocity: Symmetry, systematics diagnostic and cosmological implication

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Galaxy peculiar velocities directly probe the growth of large-scale structure, but exploiting their broader cosmological potential requires reliable measurements of galaxy motions. The stellar mass-binding energy (`massE`) catalog, the largest peculiar velocity sample to date, enables us to assess in detail how well the measured velocities trace the cosmic flow. We identify a significant dependence on local density, with systematic negative velocity offsets in denser regions. A similar trend in the SDSS Fundamental Plane samples suggests a possible common source of systematics among different distance indicators. We develop an empirical correction using the observed galaxy density as proxy, substantially reducing this trend. Although the physical origin remains uncertain, the associated bias is primarily confined to scales $\lesssim 20\,\mathrm{Mpc}/h$, leaving current analyses largely unaffected on larger scales. As an independent diagnostic, we show that rotational symmetry provides model-independent angular signatures for velocity statistics, where dipolar and quadrupolar anisotropies dominate the density-velocity cross-correlation and velocity auto-correlation, respectively. The unprecedented statistical precision of the `massE` sample enables a detection of the dipolar dependence with high significance, providing direct evidence that the measured velocities trace the underlying cosmic flow. Furthermore, we present the first cosmological analysis jointly modeling full-shape galaxy density and velocity statistics within the Lagrangian bias framework, and obtain $f\sigma_8 = 0.40^{+0.06}_{-0.05}$ at $0.02 < z < 0.2$, consistent with the Planck ΛCDM prediction. Mild discrepancies in the amplitudes of the velocity statistics complicate the cosmological interpretation, and current data cannot determine whether they arise from sample variance or measurement systematics. Altogether, these results provide practical lessons for future peculiar velocity measurements, and highlight the promise of the `massE` ruler for advancing the peculiar velocity cosmology with larger, cleaner samples over a wider redshift coverage.

I. INTRODUCTION

Galaxy peculiar velocities trace the growth of large-scale structure, serving as a cosmological probe to reveal the dark sectors, neutrino properties, and the nature of gravity [1–7]. On linear scales, guaranteed by the weak equivalence principle, galaxy peculiar velocities $\mathbf{v}(\mathbf{x})$ trace the matter density contrast $\delta_m(\mathbf{x})$ and satisfy $\nabla\cdot\mathbf{v}(\mathbf{x}) = -f\mathcal{H}\delta_m(\mathbf{x})$. The velocity statistics are therefore sensitive to the matter fluctuation amplitude σ_8 multiplied with the linear growth rate $f \equiv \mathrm{d}\ln\delta_m/\mathrm{d}\ln a$. Moreover, combining peculiar velocities with galaxy clustering and gravitational lensing, which probes the Weyl potential, enables a direct consistency test between matter motion and light deflection, and hence of the alternative theories of gravity [8–10]. Such tests have been performed using velocity information inferred from redshift space distortions [11–17], but not yet using direct velocity measurements. Realizing the full cosmological potential of peculiar velocities demands large samples of reliable velocity measurements.

Direct measurements of galaxy peculiar velocities require independent distance estimates, where the line-of-sight velocity is determined from the difference between the observed redshift and the cosmological redshift inferred by distance. Two widely used distance indicators are the Fundamental Plane (`FP`) relation for early-type galaxies [18, 19] and the Tully-Fisher (`TF`) relation for rotationally supported late-type galaxies [20, 21]. Over the past two decades, increasingly large peculiar velocity catalogs have been constructed from surveys including 6dFGS [22–24], 2MASS [25, 26], the Sloan Digital Sky Survey (SDSS) [27–29], and the Dark Energy Spectroscopic Instrument (DESI) [30–33]. The SDSS `FP` catalog provides distances and peculiar velocities for 34,059 galaxies over approximately $7000\,\mathrm{deg}^2$ [27]. The DESI survey nearly triples this measurement to 98,292 `FP` galaxies in the first release, and is expected to obtain $\sim 186{,}000$ sources using both `FP` and `TF` rulers over $14{,}000\,\mathrm{deg}^2$ in the final release [30]. Complementary distance

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indicators, including surface brightness fluctuations in early-type galaxies [34] and type Ia supernovae magnitude fluctuations [35, 36], provide additional constraints of the local velocity field [37]. The TF measurements from 21 cm observations [38] and luminosity distance fluctuations in bright sirens [39] offer further opportunities to expand peculiar velocity samples, extending the direct velocity detection to higher redshift coverage.

The recently proposed galaxy stellar mass-binding energy (massE) relation [28, 40, 41] opens a new avenue to advance the peculiar velocity cosmology. This empirical relation connects stellar mass to the combination $V_e R_e^{1/4}$, a proxy related to galaxy binding energy, where V_e is the characteristic dynamical velocity and R_e is the effective radius. It has been shown to hold across a wide variety of galaxy types and stellar masses [40], motivating its use as a distance indicator beyond the galaxy populations traditionally targeted by the FP and TF relations. For the elliptical galaxy samples considered following, the distance calibration involves only two nuisance parameters [28]. Applications to SDSS galaxies have already yielded tight constraints on both the background expansion and the growth of structure [28, 41]. These results motivate extending the massE approach to larger samples and a wider redshift range with DESI and future spectroscopic surveys.

The growing peculiar velocity samples improve the measurement precision, but realizing their full statistical power requires careful control of systematic biases. As statistical uncertainties decrease, residual biases in the inferred velocities may become a leading limitation on the cosmological analysis, and potentially mimic departures from the standard cosmological model. In this work, we therefore present a more detailed characterization of potential systematics. Specifically, we address two questions. First, *how do the measurement systematics bias the inferred peculiar velocity field?* We investigate the environmental dependence of the measured velocities, construct an empirical density-dependence correction, and use the characteristic angular relations of velocity statistics to diagnose the residual contaminations. Second, *to what extent are the measured peculiar velocities consistent with theoretical predictions?* We jointly model the galaxy density and velocity statistics using the Lagrangian bias framework, constrain the structure growth rate, and test the consistency with the ΛCDM prediction.

We use the massE peculiar velocity catalog from Ref. [28], which contains 229,890 elliptical galaxies from the SDSS main galaxy sample (MGS) over redshift $0.02 < z < 0.2$ with $6725\,\mathrm{deg}^2$ sky coverage. It is the largest peculiar velocity dataset to date, providing statistic measurements with unprecedented precision. We use the SDSS MGS value-added safe0 samples [42] from the NYU Value-Added Galaxy Catalog (NYU-VAGC) [43] as the galaxy density sample. We restrict its analysis to the contiguous region in the northern Galactic cap that overlaps the peculiar velocity catalog. The accompanying files describing the survey mask and window function are also provided therein, and we use them as inputs to the pymangle software [44] to generate random samples for the density catalog. Although the massE velocity and MGS density samples share the same parent data, the well-characterized selection function of safe0 allows us to account for survey geometry when estimating the density field and measuring the correlation functions.

II. SYMMETRY AND VELOCITY STATISTIC

We express the galaxy peculiar velocity along line-of-sight as $u = \mathcal{H}^{-1}\mathbf{v}\cdot\hat{n}$, where $\mathcal{H} = aH(z)$ is the conformal Hubble parameter and $\hat{n}$ denotes the line-of-sight direction. Here, u has dimensions of comoving length and represents the radial displacement that maps galaxy position $\mathbf{x}$ in real space and to that of $\mathbf{s}$ in redshift space, $\mathbf{s} = \mathbf{x} + u\,\hat{n}$. We adopt the Planck 2018 cosmology [45], and evaluate $\mathcal{H}$ and all subsequent model predictions at the mean redshift of the velocity sample, $z = 0.093$. We consider three summary statistics, including galaxy velocity auto-correlation, density-velocity cross-correlation, and density auto-correlation. Because velocities are sampled only at galaxy positions, the corresponding statistics are weighted by the number density of velocity tracers. These anisotropic correlation functions are therefore

$$\xi^{uu}(s,\mu) = \frac{\langle[1+\delta_v(\mathbf{s}_1)][1+\delta_v(\mathbf{s}_2)]u(\mathbf{s}_1)u(\mathbf{s}_2)\rangle}{1+\langle\delta_v(\mathbf{s}_1)\delta_v(\mathbf{s}_2)\rangle} \tag{1}$$

$$\xi^{gu}(s,\mu) = \frac{\langle[1+\delta_g(\mathbf{s}_1)][1+\delta_v(\mathbf{s}_2)]u(\mathbf{s}_2)\rangle}{1+\langle\delta_g(\mathbf{s}_1)\delta_v(\mathbf{s}_2)\rangle} \tag{2}$$

$$\xi^{gg}(s,\mu) = \langle\delta_g(\mathbf{s}_1)\delta_g(\mathbf{s}_2)\rangle \tag{3}$$

where $\mathbf{s} = \mathbf{s}_2 - \mathbf{s}_1$, $s = |\mathbf{s}|$ and $\mu = \hat{n}\cdot\mathbf{s}/s$. We denote the overdensity of the velocity sample as $\delta_v(\mathbf{s})$, distinguishing it from the overdensity $\delta_g(\mathbf{s})$ of the galaxy density tracers. In practice, we estimate ξ^{uu} by averaging products of line-of-sight velocities over tracer pairs, and measure ξ^{gu} as the mean velocity around density tracers with subtraction of the corresponding mean around random points. The density auto-correlation ξ^{gg} is measured using the Landy-Szalay estimator. On large scales $s \gtrsim 10\,\mathrm{Mpc}/h$, two velocity statistics are well approximated by their leading-order contributions, with $\xi^{uu} \simeq \langle u_1 u_2\rangle$ and $\xi^{gu} \simeq \langle\delta_1 u_2\rangle$.

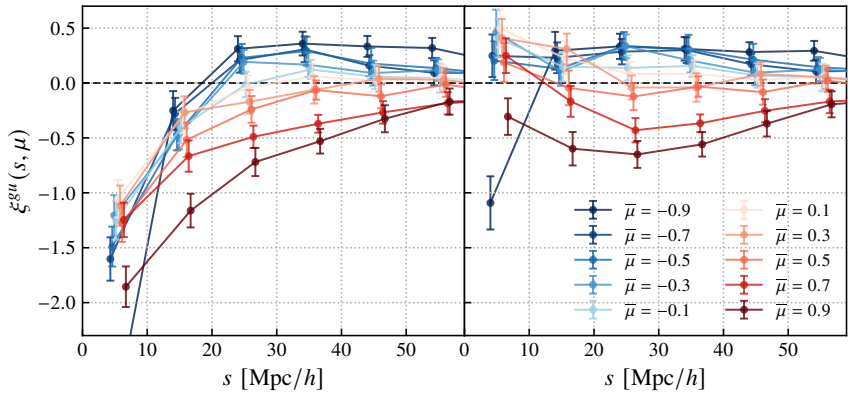


FIG. 1. **Galaxy density-velocity cross-correlation function measured from the massE and MGS catalogs.** Curves with different colors denote measurements in different angular bins, measured with bin sizes of $\Delta\mu = 0.1$ and $\Delta s = 10\,\mathrm{Mpc}/h$. **Left panel**: Raw results without correction. The uncorrected measurements become increasingly negative toward smaller separations, indicating more negative peculiar velocities in denser environments. **Right panel**: Results after applying the density-dependence correction. The expected antisymmetry about $\mu = 0$ is largely restored on scales $s \gtrsim 10\,\mathrm{Mpc}/h$.

In the plane-parallel approximation, statistics of density field contain only even-parity modes, where the anisotropic components arise from redshift space distortions. By contrast, the peculiar velocity field produces an odd-parity statistic when cross-correlating with galaxy density, dominated by the dipole, while its leading anisotropy in auto-correlation is dominated by the quadrupole. These distinctive signatures separate the velocity from other scalar observables, and enables a direct diagnostic of peculiar velocity measurements. In real space, statistical isotropy requires the mean velocity around a density tracer to be aligned with the pair separation, and the isotropic velocity correlation tensor contains only components proportional to the identity tensor and the outer product of the unit pair direction. Consequently, two velocity statistics can be decomposed as

$$\xi^{gu}(r,\mu_r) = \xi_1^{gu}(r)\mathcal{L}_1(\mu_r) \tag{4}$$
$$\xi^{uu}(r,\mu_r) = \xi_0^{uu}(r)\mathcal{L}_0(\mu_r) + \xi_2^{uu}(r)\mathcal{L}_2(\mu_r) \tag{5}$$

Here, $\mathbf{r} = \mathbf{x}_2 - \mathbf{x}_1$, $r = |\mathbf{r}|$ and $\mu_r = \hat{n}\cdot\mathbf{r}/r$. The correlation function multipoles are given by $\xi_\ell(r) = \frac{2\ell+1}{2}\int_{-1}^{1} d\mu\,\mathcal{L}_\ell(\mu)\xi(r,\mu)$, where $\mathcal{L}_\ell(\mu)$ is Legendre polynomial. These angular structures follow from rotational symmetry and the vector nature of the velocity field, without requiring a specific theory of gravity or cosmological model. It therefore enables a model-independent diagnostic for the peculiar velocity field. However, two observational effects, redshift space distortions [46, 47] and wide angle effects [48, 49], modify these simple angular structures by generating additional components. Redshift space distortions source higher order multipoles of velocity statistics with the parity preserved, whereas wide angle effects can additionally produce opposite-parity components. Nevertheless, on linear scales, redshift space distortions do not contribute to ξ^{uu}, while primarily generate an $\ell = 3$ component in ξ^{gu} with amplitude suppressed by a factor of $\sim \frac{2}{5}f/\left(1+\frac{3}{5}f\right) \sim 0.15$ relative to the leading-order dipole. Wide angle effect becomes significant only for large angular scale, typically at separation $\gtrsim 100\,\mathrm{Mpc}/h$ over our redshift coverage, remaining to be dominated by cosmic variance. As we show below, these effects are subdominant to current observational uncertainties. Therefore, even in realistic observations, Eqs. (4) and (5) still serve as good approximations to capture the leading-order angular dependence of the velocity statistics.

III. DENSITY-DEPENDENCE OF THE PECULIAR VELOCITY RESIDUAL

Before examining the angular dependence, we inspect the full galaxy density-velocity cross-correlation. The left panel of Fig. 1 shows the measured $\xi^{gu}(s,\mu)$ as a function of separation s for different μ bins. At leading order, Eq. (4) predicts a dipolar, odd-parity signal satisfying $\xi^{gu}(s,-\mu) = -\xi^{gu}(s,\mu)$. The measurements approximately recover this antisymmetry at large separations but depart from it on small scales $s \lesssim 20\,\mathrm{Mpc}/h$. In particular, the measurements are systematically negative across all μ bins on $s \simeq 15\,\mathrm{Mpc}/h$, and the excess becomes even more pronounced at $s \simeq 5\,\mathrm{Mpc}/h$. Such significantly negative contribution is not expected from the cosmic velocity field, and indicates a systematic correlated with the galaxy distribution.

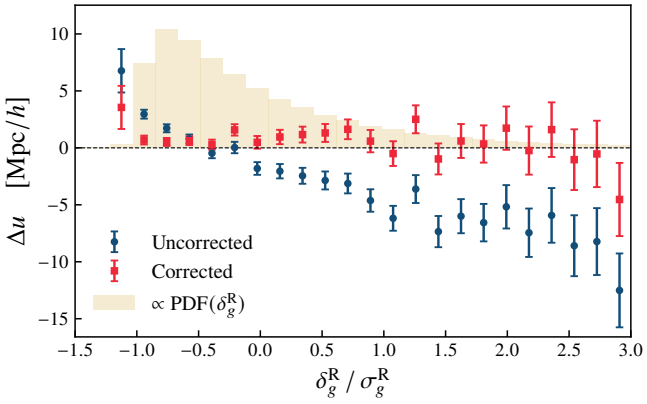


FIG. 2. **Density dependence of the measured `massE` peculiar velocities.** Blue circles and red squares denote the residual mean Δu of peculiar velocities in bins of the galaxy overdensity $\delta_g^{\rm R}$, measured before and after the density-dependence correction, respectively. Here, $\delta_g^{\rm R}$ is obtained by smoothing the galaxy density field with a Gaussian kernel of width $R = 2\,{\rm Mpc}/h$. The yellow shaded histogram shows $\delta_g^{\rm R}$ distribution scaled by $\sigma_g^{\rm R} = \sqrt{\langle(\delta_g^{\rm R})^2\rangle}$. Without correction, Δu decreases systematically with increasing density, indicating a significant environmental dependence in the measured peculiar velocities. After applying the correction, this trend is largely removed.

To investigate this environmental dependence, we measure the mean peculiar velocity as function of local galaxy number density. We construct a galaxy overdensity field from the MGS galaxies and corresponding random samples, and smooth it with a Gaussian kernel of width $R = 2\,{\rm Mpc}/h$ to suppress shot noise. Each velocity tracer is then assigned the smoothed overdensity at its observed position. Fig. 2 present the conditional mean velocity in each overdensity bin. If the measurements consisted only the underlying peculiar motion and fluctuations uncorrelated with the cosmic environment, this conditional mean should vanish. However, we observe a clear correlation between the mean velocity and local density, indicating that the measurements are biased by local environment. The samples in overdense regions exhibit preferentially negative measured velocities, consistent with the negative excess in ξ^{gu}. This observed trend persists when varying the density smoothing scale or selecting different redshift ranges, suggesting that it is not specific to a particular density definition or redshift selection.

We therefore use the galaxy overdensity field to construct empirical templates for this environment dependence, and validate whether it captures the systematic trend. To capture contributions from different spatial scales, we construct $N_\delta = 12$ filtered density fields by $\delta_g^\alpha(\mathbf{k}) \equiv W_\alpha(k)\delta_g(\mathbf{k})$ in Fourier space, using Gaussian band-pass filters $W_\alpha(k) = \exp\left[-\frac{(\ln k - \ln k_\alpha)^2}{2\sigma_W^2}\right]$. We adopt a bandwidth $\sigma_W = 0.3$ and choose the characteristic scales k_α to be uniformly spaced in $\log_{10}\left[k_\alpha/({\rm Mpc}^{-1}h)\right]$ over $[-1.8, 1]$. For each filtered density field δ_g^α, we group the velocity tracers into bins of local filtered overdensity, and measure the mean velocity Δu_a^α for each bin, where a index the bin. This mean velocity is then assigned to every tracer in the corresponding bin, defining the template

$$\Delta u^\alpha(\mathbf{s}_i) \equiv \Delta u_a^\alpha \qquad i \in \mathcal{B}_a^\alpha \ , \tag{6}$$

for i-th tracer observed at $\mathbf{s}_i$, where $\mathcal{B}_a^\alpha$ denotes the set of tracers in the a-th overdensity bin of the α-th filtered field. Each template therefore characterizes the observed mean velocity as a function of the local density on a particular spatial scale. The density-dependent contamination is modeled as a linear combination of these templates,

$$\Delta\hat{u}(\mathbf{s}_i) = \sum_\alpha \mathcal{W}_\alpha\, \Delta u^\alpha(\mathbf{s}_i) \ , \tag{7}$$

and it is subtracted from the measured velocity to obtain the corrected velocity

$$\hat{u}_{\rm c}(\mathbf{s}_i) = \hat{u}(\mathbf{s}_i) - \Delta\hat{u}(\mathbf{s}_i) \ . \tag{8}$$

Here, the coefficients $\mathcal{W}_\alpha$ are determined jointly, using the expectation that the conditional mean of $\hat{u}_{\rm c}$ vanishes in all overdensity bins across all filtered fields, by maximizing the Gaussian likelihood of the binned residuals (details in Appendix B).

This correction approach substantially reduces the observed environmental trend. As shown by the red squares in Fig. 2, the conditional mean velocity becomes broadly consistent with zero after the density-dependence correction.

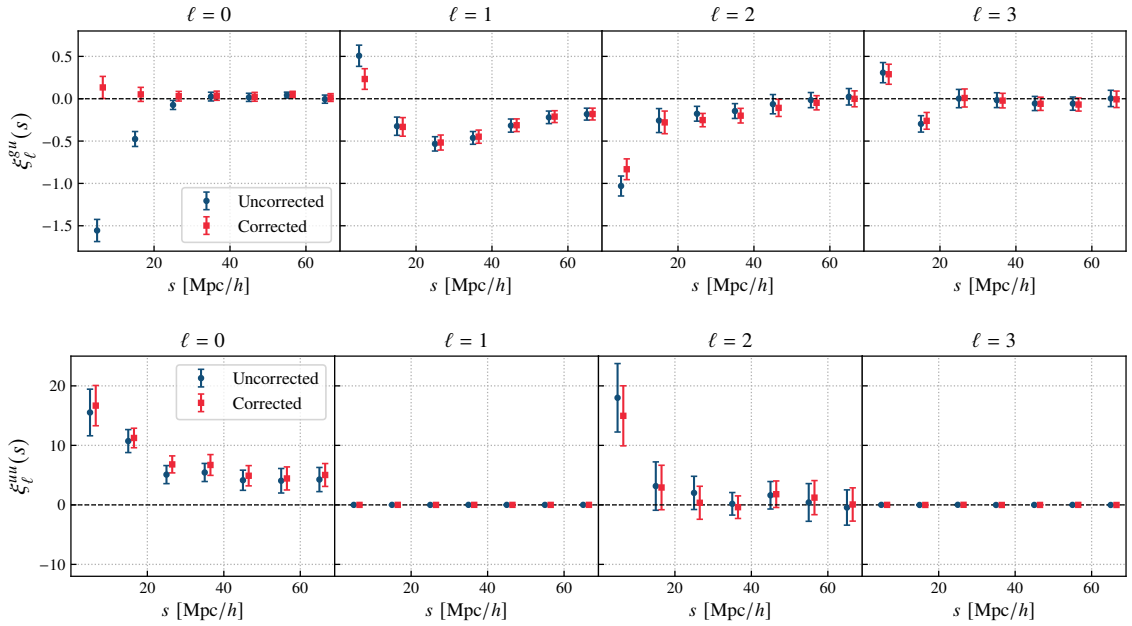


FIG. 3. **Multipoles of the galaxy density-velocity cross-correlation $\xi_\ell^{gu}(s)$ (top panels) and velocity auto-correlation $\xi_\ell^{uu}(s)$ (bottom panels)**, measured using the `massE` velocity catalog and MGS density catalog. Columns from left to right show $\ell = 0, 1, 2$ and 3. Blue circles and red squares denote measurements before and after the density-dependence correction, respectively. The odd multipoles ξ_1^{uu} and ξ_3^{uu} vanish because of the symmetry of the auto-correlation estimator.

The right panel of Fig. 1 also shows a substantial reduction in the excess negative contribution to ξ^{gu}, largely restoring the expected antisymmetry about $\mu = 0$. This improvement provides a complementary consistency check using a two-point statistic that is not explicitly fitted by the correction procedure. Fig. 3 further compares the velocity correlation function multipoles ξ_ℓ^{gu} and ξ_ℓ^{uu} before and after the correction. A nonzero ξ_3^{gu} can arise from redshift space distortions and need not indicate contamination. In contrast, the residual small-scale ξ_2^{gu} indicates an anisotropic systematic contribution, since redshift space distortions preserve the odd parity on these scales. One possible source of the residual quadrupole is that the density-dependent trends rely on the intrinsic galaxy density, while the templates are constructed from observed galaxy distribution, where redshift space distortions contaminate the local density estimate on small scales. It also partially explains the residual deviation remaining at the smallest separation bins $s \simeq 5\,\mathrm{Mpc}/h$, shown in Fig. 1. Additional treatment of this redshift-space anisotropy may further improve the correction accuracy. Nevertheless, the substantial reduction of the density-dependent trends in both one-point and two-point statistics identifies local galaxy density as an empirical proxy of the systematic residuals.

The physical origin of this density dependence remains uncertain. One possible explanation is an environment-dependent mismatch between the measured stellar velocity dispersion and the dynamical velocity entering the `massE` relation. This `massE` catalog uses elliptical galaxies selected primarily by morphology, with their dynamical velocities estimated from stellar velocity dispersions measured through fiber spectra [28]. These measurements do not fully characterize the rotational support of galaxies. If the fraction of rotational component in galaxy dynamics were independent of environment, they would primarily contribute an additional scatter or a calibration offset. However, galaxy rotational support correlates with stellar population properties and environment, where galaxies in less dense environments tending to exhibit more rotation [50–52]. The neglect of rotation measurement, together with possible contamination by spiral galaxies, could therefore contribute to the environmental dependence of the inferred peculiar velocities. Testing this hypothesis requires spatially resolved spectroscopic observations, which we leave to future work.

Similar environmental trends also occur in peculiar velocities constructed from the `FP` relation. In the SDSS peculiar velocity analysis [27], fitting a single `FP` relation to the full sample produces a residual correlated with group richness, where richer groups preferentially exhibit negative peculiar velocities. This group-richness dependence is also identified in the DESI DR1 `FP` samples [19]. Although the corresponding bias is reduced by fitting the `FP` relation separately to group-richness subsamples, the origin of the trend remains unclear. Because group richness and local galaxy density both trace the large-scale clustering, we hypothesize that the two observed dependencies share a common origin. To test this possibility, we apply our density-dependence correction to the SDSS `FP` catalog. Fig. 4 present

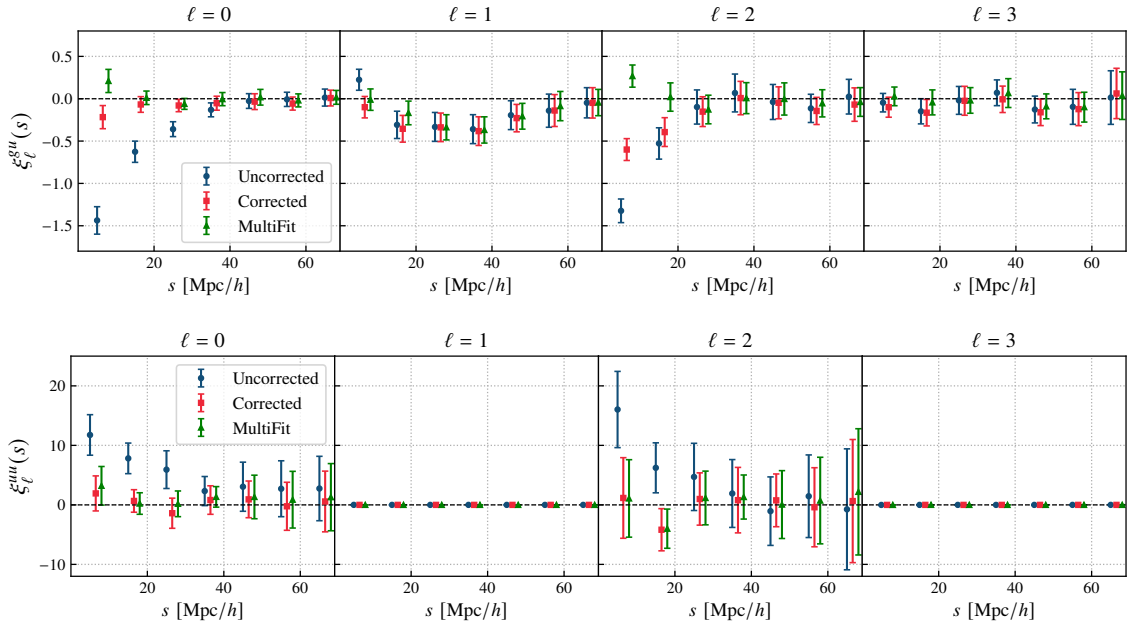


FIG. 4. Similar to Fig. 3, but measured from **the SDSS Fundamental Plane peculiar velocity catalog** [27] and MGS density catalog. Blue circles denote the uncorrected measurements, and red squares show the results after applying the density-dependence correction. Green triangles indicate the peculiar velocities obtained by fitting the `FP` relation separately to different group-richness subsamples (Section 5.3 of Ref. [27]).

the velocity statistics from the SDSS `FP` samples, comparing the multipoles obtained from a single `FP` fit, from the same catalog after our density-dependence correction, and from separate `FP` fits in group-richness bins [27]. The two correction procedures yield broadly similar results, particularly for the ξ_ℓ^{uu}. Multiple `FP` fits reduce the residual ξ_2^{gu} more effectively than the density-based correction, indicating that group membership better captures the intrinsic environment in redshift space. For the leading cosmological signals, particularly ξ_1^{gu}, the `massE` measurements are substantially more precise than the `FP` measurements, given the larger `massE` sample size. Nevertheless, the similar response of the `massE` and `FP` measurements to environmental corrections suggests that the two distance indicators may share an underlying source of the systematic effect. Current data are insufficient either to further test this hypothesis, or to determine whether the observed variations are physical or arise from observational systematics. Further investigation with larger samples is needed to clarify these origin.

IV. DIAGNOSTIC OF PECULIAR VELOCITY WITH ANGULAR RELATIONS

As discussed in Sec. II, the density-velocity cross-correlation is expected to be dominated by a dipolar signature, while the leading anisotropy of the velocity auto-correlation is quadrupolar. We can therefore assess the fidelity of the measured peculiar velocity field by testing the angular relations

$$\xi^{gu}(s,\mu) \propto \mu \,, \tag{9}$$

$$\xi^{uu}(s,\mu) - \xi_0^{uu}(s) \propto \mu^2 - \frac{1}{3} \,. \tag{10}$$

These leading-order relations follow from the rotational symmetry and the vector nature of the velocity field, without assuming a specific cosmological model. Residual environmental dependence, selection effects, and secondary trends in galaxy scaling relations, do not generally reproduce these characteristic angular dependences. Agreement with Eqs. (9) and (10) therefore provides a direct, data-driven test of whether fluctuations in the distance residuals predominantly trace the underlying peculiar velocity field.

Before applying this test to the data, we quantify how redshift space distortions and wide angle geometry modify the idealized relations of Eqs. (9) and (10). We construct mock samples from the dark matter snapshot of the CosmicGrowth-6620 simulation [53]. We place an observer at a fixed position, convert the comoving distance of each matter particle into a cosmological redshift, and include the redshift space distortion along the local line-of-sight.

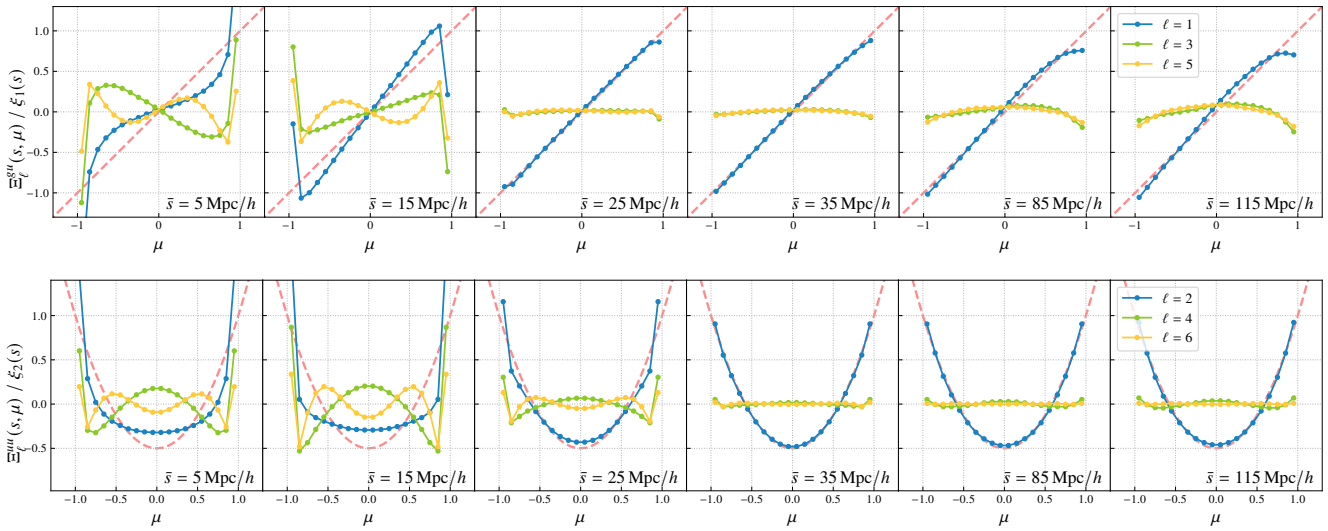


FIG. 5. **Validation of the angular relations $\xi^{gu}(s,\mu) \propto \mu$ and $\xi^{uu}(s,\mu) - \xi^{uu}_0(s) \propto \mu^2 - \frac{1}{3}$ using N-body simulation.** Each panel shows the residual correlation functions of Eq. (11) for ξ^{gu} (**top panels**) or Eq. (12) for ξ^{uu} (**bottom panels**), measured from dark matter mocks including redshift space distortions and wide angle effects, and normalized by the leading-order anisotropic multipole. Panels from left to right show results in different s bins, and solid lines of different colors denote different multipole orders ℓ. The red dashed lines indicate the theoretical expectations of the leading angular relations. At separations $\gtrsim 20\,\mathrm{Mpc}/h$, the results agree well with the expected angular relation, while the higher-order residuals are closed to zero at $\sim 10\%$ level. At smaller separations, significant deviations emerge because of nonlinear redshift space distortions.

Then, we select particles within $0.02 < z < 0.2$, and downsample them separately to match the effective redshift distributions of the density and velocity tracers, respectively. The resulting mocks incorporate both redshift space distortions and wide angle geometry. We measure the velocity statistics using the same estimators as for the survey data. To isolate departures from the leading angular structure, we perform the Gram-Schmidt orthogonalization, and define the residual correlation functions

$$\Xi^{gu}_\ell(s,\mu) \equiv \xi^{gu}(s,\mu) - \sum_{\ell'=1,3,\cdots}^{\ell'<\ell} \xi^{gu}_{\ell'}(s)\mathcal{L}_{\ell'}(\mu) \tag{11}$$

$$\Xi^{uu}_\ell(s,\mu) \equiv \xi^{uu}(s,\mu) - \sum_{\ell'=0,2,\cdots}^{\ell'<\ell} \xi^{uu}_{\ell'}(s)\mathcal{L}_{\ell'}(\mu) \tag{12}$$

Here, the orthogonalization removes lower-order multipoles of the matched parity while retaining all components of the opposite parity. For a pure dipole, $\Xi^{gu}_1/\xi^{gu}_1 = \mu$, and $\Xi^{gu}_L = 0$ for odd $L \geq 3$. For an auto-correlation containing only monopole and quadrupole, $\Xi^{uu}_2/\xi^{uu}_2 = \frac{3}{2}\left(\mu^2 - \frac{1}{3}\right)$, and $\Xi^{uu}_L = 0$ for even $L \geq 4$. Fig. 5 presents these residual correlation functions measured from the dark matter mocks. The density-velocity cross-correlation closely follows the dipolar expectation on $\gtrsim 20\,\mathrm{Mpc}/h$, with higher-order residuals remaining at the $\lesssim 10\%$ level for very large separations. At smaller separations, the deviations become more pronounced. The dipolar approximation remains valid for pairs oriented away from the line-of-sight in bin $s \simeq 15\,\mathrm{Mpc}/h$, but breaks down at $s \simeq 5\,\mathrm{Mpc}/h$, because of nonlinear redshift space distortions arising from virial motions within halos. The velocity auto-correlation behaves similarly. Its angular structure is well described by a quadrupole on large scales, while exhibits a pronounced U-shaped profile that differs from a pure quadrupole in smaller separation bin. These characteristic angular dependences also hold for simulated galaxies in hydrodynamical simulations across different populations (e.g., Fig. 21 & 22 shown in Appendix), because they follow from the rotational symmetry rather than the clustering properties of the tracers. Thus, over the scale range $20 \lesssim s \lesssim 100\,\mathrm{Mpc}/h$, the angular relations of Eqs. (9) and (10) remain robust signatures of the peculiar velocity field despite two observational effects.

Fig. 6 presents the angular relations of the velocity statistics measured from the `massE` catalog. For the density-velocity cross-correlation, measurements at $s \gtrsim 20\,\mathrm{Mpc}/h$ closely follows the relation $\xi^{gu}(s,\mu) \propto \mu$, consistent with the dipolar dependence. Before the density-dependence correction, the $s \simeq 5\,\mathrm{Mpc}/h$ and $s \simeq 15\,\mathrm{Mpc}/h$ bins deviate significantly from the expected antisymmetry, indicating substantial systematics. After the correction, the $s \simeq 15\,\mathrm{Mpc}/h$ bin broadly follows the dipolar angular dependence, while the $s \simeq 5\,\mathrm{Mpc}/h$ bin retains a substantial

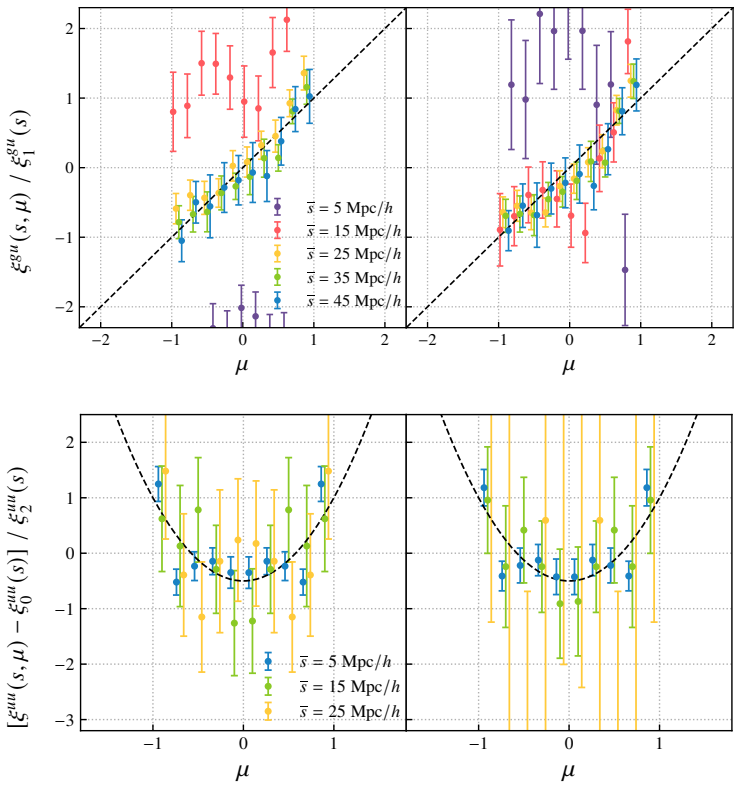


FIG. 6. **Angular relations of the measured velocity statistics of the `massE` catalog, before (left panels) and after (right panels) applying the density-dependence correction.** **Top panels**: galaxy density-velocity cross-correlation $\xi^{gu}(s,\mu)$, same results presented in Fig. 1 but normalized by the overall amplitude within each s bin. On scales $\gtrsim 20\,\mathrm{Mpc}/h$, the measurements exhibit excellent agreement with the expected angular scaling, i.e., $\xi^{gu}(s,\mu) \propto \mu$. After correction for density-dependence, the measurements on $\sim 10\,\mathrm{Mpc}/h$ also broadly follow the theoretical expectation. **Bottom panels**: velocity auto-correlation $\xi^{uu}(s,\mu)$ with normalization ξ_2^{uu}. In both the uncorrected and corrected measurements, the expected angular dependence $\xi^{uu}(s,\mu) - \xi_0^{uu}(s) \propto \mu^2 - \frac{1}{3}$ is not discernible due to the low detection significance of individual bins.

departure. The velocity auto-correlation is detected at substantially lower significance, and its angular dependence cannot be resolved outside the smallest-separation bin. At $s \simeq 5\,\mathrm{Mpc}/h$, it exhibits a characteristic U-shaped profile, qualitatively resembling the nonlinear behavior in simulations. These results indicate that the `massE` galaxy peculiar velocities reliably trace the underlying cosmic flow on scales $\gtrsim 10\,\mathrm{Mpc}/h$, and the density-dependence correction approach effectively mitigates the systematic bias.

V. STRUCTURE GROWTH RATE FROM GALAXY PECULIAR VELOCITY

The preceding sections use the intrinsic angular relations of the velocity statistics to assess the measured peculiar velocity field. In the following, we perform a full-shape modeling of the broadband features of the velocity statistics, jointly with the galaxy density auto-correlation, to constrain $f\sigma_8$, and test the consistency of the measured peculiar velocities with the ΛCDM prediction. For the cosmological analysis, we use the velocity samples without applying the empirical density-dependence correction, and restrict the analysis to large scales, where the contamination is largely suppressed.

A. Nonlinear modeling of galaxy statistics

We describe galaxy clustering using the Lagrangian bias expansion and model its nonlinear evolution with Lagrangian perturbation theory [47, 54–66]. This framework consistently predicts three statistics of ξ^{gg}, ξ^{gu} and ξ^{uu} in redshift space. In Lagrangian space of coordinate $\mathbf{q}$, the galaxy overdensity field is expanded schematically as series

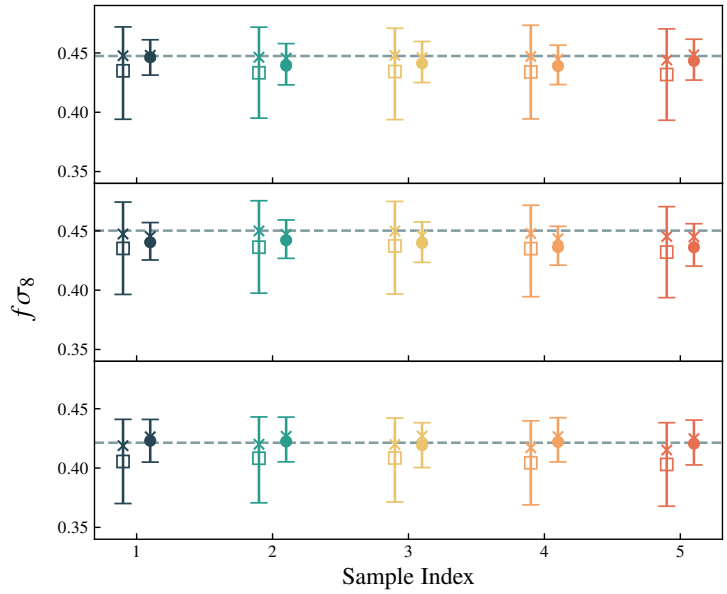


FIG. 7. **Validation of the Lagrangian bias model for velocity statistics using the FLAMINGO simulations**: L1_m9 (**top panel**), Planck (**middle panel**) and LS8 (**bottom panel**). Error bars show the $1\,\sigma$ marginalized constraints on $f\sigma_8$ from the analysis with peculiar velocity alone (open squares) and the joint analysis with both galaxy density and peculiar velocity (filled circles). Crosses mark the MAP estimate values, and their offsets from the marginalized means reflect the non-Gaussianity of the posterior distribution. From left to right, different colors denote results from different galaxy samples selected by stellar mass. Across all three cosmologies and galaxy samples with different clustering biases, the theoretical model consistently recovers the truth values of $f\sigma_8$ (gray dashed lines), validating the theoretical framework used to interpret the data.

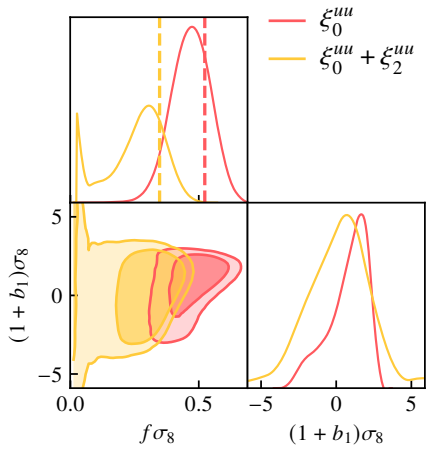


FIG. 8. **Posterior distributions of the model parameters from the `massE` velocity statistics.** Red and yellow contours show the results obtained using data vector $\{\xi_0^{uu}\}$ and $\{\xi_0^{uu}, \xi_2^{uu}\}$, respectively, and dashed vertical lines mark their corresponding MAP estimates of $f\sigma_8$ values. Including ξ_2^{uu} substantially shifts the inferred $f\sigma_8$ toward zero.

of operators as

$$1 + \delta_g^L(\mathbf{q}) = 1 + b_1\delta(\mathbf{q}) + b_2\delta^2(\mathbf{q}) + b_s s^2(\mathbf{q}) + \mathcal{O}(\delta^3)\,. \tag{13}$$

The equivalence principle and rotational symmetry dictates that the operators building the galaxy overdensity field should be Euclidean scalars constructed from the second order spatial derivatives of the gravitational potentials Φ, and their coefficients $\{b_1, b_2, b_s, \cdots\}$ are galaxy bias [67]. Here, $\delta(\mathbf{q}) \equiv \nabla^2\Phi$ is the matter density contrast, $s^2 \equiv s_{ij}s_{ij}$ is the scalar tidal operator, and $s_{ij} = \left(\partial_i\partial_j\,\nabla^{-2} - \frac{1}{3}\delta_{ij}^K\right)\delta$ is the tidal tensor. The number conservation maps the

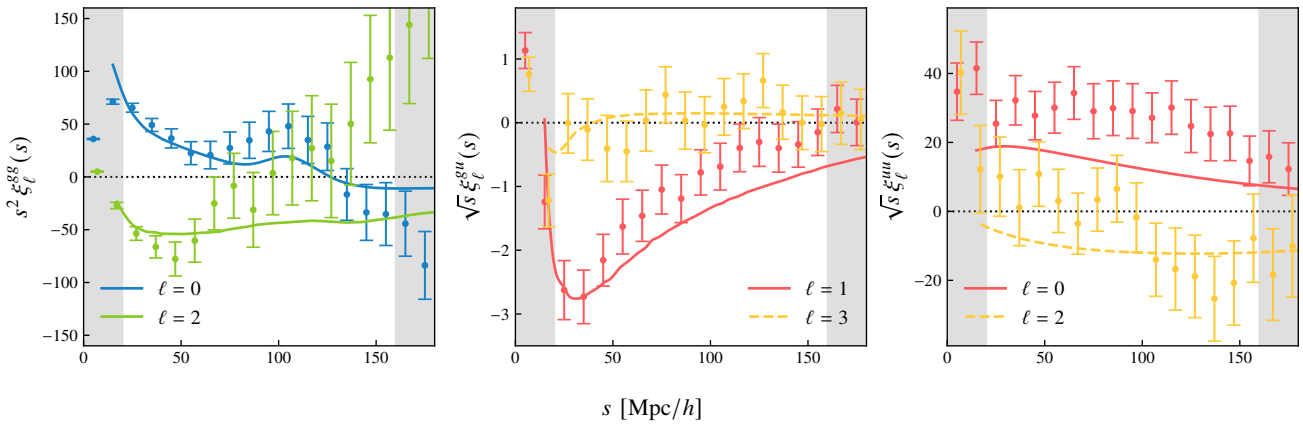


FIG. 9. **Galaxy statistics measured from the `massE` velocity and MGS density catalogs.** Panels from left to right show the multipoles of the galaxy density auto-correlation $\xi^{gg}_{0,2}$, the density-velocity cross-correlation $\xi^{gu}_{1,3}$ and the velocity auto-correlation $\xi^{uu}_{0,2}$. Circles with error bars denote the measurements. Solid curves show the Lagrangian bias expansion model evaluated at the MAP parameters with $f\sigma_8 = 0.408$, obtained by jointly fitting the fiducial data vector $\{\xi^{gg}_{0,2}, \xi^{gu}_1, \xi^{uu}_0\}$. Dashed curves show the corresponding predictions for ξ^{gu}_3 and ξ^{uu}_2, which are not included in the fit. Gray shaded regions indicate the scales excluded during the model fitting.

galaxy density field from Lagrangian coordinates $\mathbf{q}$ to redshift-space coordinates $\mathbf{s}$ according to

$$1 + \delta_g(\mathbf{s}) = \int \mathrm{d}^3\mathbf{q}\, \left[1 + \delta_g^L(\mathbf{q})\right]\, \delta^D(\mathbf{s} - \mathbf{q} - \mathbf{\Psi}^s(\mathbf{q}, z))\,, \tag{14}$$

where $\mathbf{\Psi}$ is the displacement field and $\mathbf{\Psi}^s = \mathbf{\Psi} + u\hat{n}$ is its redshift-space counterpart. The peculiar velocity field is related to the displacement through $\mathbf{v} = a\dot{\mathbf{\Psi}}$, and the line-of-sight velocity becomes $u = \hat{n}\cdot\dot{\mathbf{\Psi}}/H$. In Fourier space, we can therefore write the two-point statistics of galaxy velocity and density fields as

$$P^s_{\mathrm{XY}}(\mathbf{k}) = \int \mathrm{d}^3\mathbf{q}\, e^{i\mathbf{k}\cdot\mathbf{q}}\, \mathcal{M}_{\mathrm{XY}}(\mathbf{q}, \mathbf{k}) \tag{15}$$

$$\mathcal{M}_{gg} = \langle\, \left[1 + \delta_g^L(\mathbf{q}_1)\right]\, \left[1 + \delta_g^L(\mathbf{q}_2)\right]\, e^{i\mathbf{k}\cdot\Delta\mathbf{\Psi}^s}\,\rangle \tag{16}$$

$$\mathcal{M}_{gu} = \langle\, \left[1 + \delta_v^L(\mathbf{q}_1)\right]\, \left[1 + \delta_g^L(\mathbf{q}_2)\right]\, u(\mathbf{q}_1) e^{i\mathbf{k}\cdot\Delta\mathbf{\Psi}^s}\,\rangle \tag{17}$$

$$\mathcal{M}_{uu} = \langle\, \left[1 + \delta_v^L(\mathbf{q}_1)\right]\, \left[1 + \delta_v^L(\mathbf{q}_2)\right]\, u(\mathbf{q}_1) u(\mathbf{q}_2)\, e^{i\mathbf{k}\cdot\Delta\mathbf{\Psi}^s}\,\rangle \tag{18}$$

where $\mathbf{q} = \mathbf{q}_2 - \mathbf{q}_1$ and $\Delta\mathbf{\Psi}^s = \mathbf{\Psi}^s_2 - \mathbf{\Psi}^s_1$. The factor $1 + \delta_v^L$ denotes the Lagrangian number density of velocity tracers. Plugging Eq. (13) into Eqs. (15)-(18), we obtain the redshift-space power spectra P^s_{gg}, P^s_{gu} and P^s_{uu}. Fourier transformation then yields the redshift-space correlation functions ξ^{gg}, ξ^{gu} and ξ^{uu}. Furthermore, the structure growth rate is constrained primarily by the overall amplitudes of the galaxy statistics, e.g., $\xi^{gu}_1 \propto f\sigma_8(b^E\sigma_8 + \frac{3}{5}f\sigma_8)$ and $\xi^{uu}_0 \propto (f\sigma_8)^2$ at the leading order, where $b^E \equiv 1 + b_1$ is the linear Eulerian bias for density tracers. Terms involving δ_v contribute mainly through nonlinear coupling on small scales, leaving the associated bias parameters poorly constrained. We therefore set $\delta_v(\mathbf{s}) = \delta_g(\mathbf{s})$ to reduce the parameter space. This approximation neglects the difference in galaxy clustering induced by velocity sample selection, but does not materially restrict the model at the precision of the current data.

For the galaxy density power spectrum P^s_{gg}, we adopt the well-established one-loop formulas from Refs. [58–61]. Here, the displacement and the velocity fields are expanded perturbatively as $\mathbf{\Psi} = \sum_m \mathbf{\Psi}^{(m)}$ and $u = \sum_m u^{(m)}$, where $u^{(m)} = fm\mathbf{\Psi}^{(m)}\cdot\hat{n}$. The leading long-wavelength contributions from $\mathbf{\Psi}^{(1)}$ and $u^{(1)}$ are resummed, retained in the exponent of the displacement mapping, while the remaining higher order contributions are expanded in the prefactor. For the density-velocity cross power spectrum P^s_{gu}, we use the redshift-space density power spectrum as a generating function for pairwise velocity statistics [47, 66]. We obtain P^s_{gu} by numerically differentiating P^s_{gg} with respect to the linear growth rate f. Because the velocity auto power spectrum P^s_{uu} is not a pairwise velocity statistic, we instead derive it directly using the same formalism as for P^s_{gg}. The resulting expressions are presented in Appendix D 2. We perform all numerical calculations with the public `velocileptors` package [68] , which we extend to include the velocity auto-correlation.

The resulting model parameters are $\{f\sigma_8, b_1\}$ for velocity-only analysis, $\{f\sigma_8, b_1, b_2, b_s\}$ for density-only analysis, and $\{f\sigma_8, b_1, b_2, b_s, \alpha_v\}$ for joint analysis of velocity and density fields. Here, α_v comes from the counterterm that

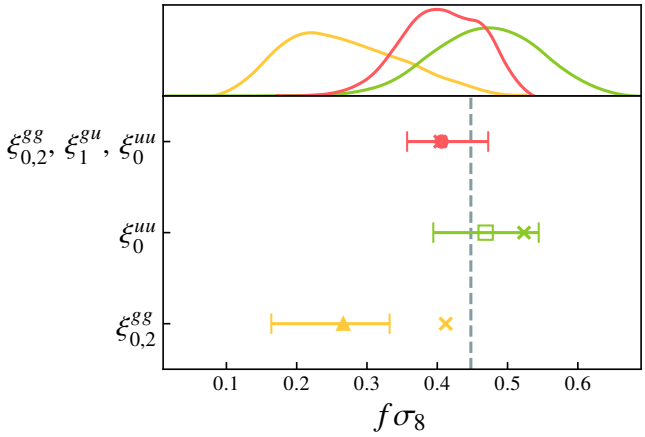


FIG. 10. **Constraints on the structure growth rate $f\sigma_8$ from the `massE` velocity and MGS density catalogs.** Results are obtained from the joint data vector $\{\xi^{gg}_{0,2}, \xi^{gu}_{1}, \xi^{uu}_{0}\}$ (**top**, red circle), the velocity auto-correlation monopole ξ^{uu}_{0} alone (**middle**, green open square), and the galaxy density multipoles $\xi^{gg}_{0,2}$ alone (**bottom**, yellow triangle). Error bars show the $1\,\sigma$ marginalized constraints, crosses mark the MAP estimates, and the curves in the upper panel show the corresponding marginalized posterior distributions. The gray dashed line indicates the prediction of the Planck ΛCDM cosmology. The density-only analysis shows a substantial difference between the marginalized and MAP estimates, reflecting prior volume effects, while including the velocity statistics mitigates these effects and yields a more robust constraint.

accounts for unresolved small-scale contributions in the density-velocity cross correlation. For the velocity auto-correlation, we retain only the linear bias because the higher order terms vanish at one-loop order. Our preliminary data vector is $\{\xi^{gg}_{0}, \xi^{gg}_{2}, \xi^{gu}_{1}, \xi^{uu}_{0}, \xi^{uu}_{2}\}$. We fit each multipole over $s \in [20, 160]\,\mathrm{Mpc}/h$ using 14 uniform separation bins. The lower limit avoids the density-dependent bias discussed above, while the upper limit follows the SDSS analysis of redshift space distortions [69].

We validate the model across different cosmologies and galaxy populations using the FLAMINGO simulations [70–72], a suite of hydrodynamical simulations designed for large-scale structure study. We consider three cosmological variants, i.e., the `L1_m9`, `Planck`, and `LS8` simulations, of which the structure growth rates are $f\sigma_8 = 0.446$, 0.448 and 0.420, respectively, at redshift $z = 0.1$. In each simulation, we treat all luminous subhalos from the SOAP catalogs [73–75] as galaxies, and construct 5 subsamples by binning their stellar mass, each with a fixed number density matching that of the MGS catalog. To mimic observational uncertainties, we add the mean velocity dispersion of the `massE` catalog to the simulated peculiar velocities as additional white noise. Rather than constructing mocks that reproduce properties of observational samples, this setup aims to test the model across a broad range of cosmologies, galaxy biases, and stochasticities. Fig. 7 summarizes the resulting marginalized constraints and the global maximum a posteriori (MAP) estimates of the $f\sigma_8$. Across all 3 cosmologies and all 5 stellar-mass samples, the inferred values are consistent with the corresponding true values within the statistical uncertainties. The velocity only analyses of $\{\xi^{uu}_{0,2}\}$ present unbiased inferences of $f\sigma_8$, while the joint analyses including the density field of $\{\xi^{uu}_{0,2}, \xi^{gu}_{1}, \xi^{gg}_{0,2}\}$ yields tighter constraints. These unbiased results across different cosmologies and tracer populations demonstrate that the theoretical model robustly recovers the underlying structure growth rate.

B. Constraints on the structure growth rate

We next fit the model to the galaxy statistics measured from the survey data. Fig. 8 compares the velocity-only analyses using ξ^{uu}_{0} alone and the combination $\{\xi^{uu}_{0}, \xi^{uu}_{2}\}$, where the monopole and quadrupole are both the leading order components in velocity auto-correlation. Fitting ξ^{uu}_{0} alone yields $f\sigma_8 = 0.47^{+0.08}_{-0.07}$, consistent with the power spectrum analysis of Ref. [28], where the small difference likely reflects the different scales probed by configuration and Fourier space statistics. Including ξ^{uu}_{2} shifts the inferred $f\sigma_8$ substantially downward and weakens, rather than improves, the constraining power relative to using ξ^{uu}_{0} alone. After including ξ^{uu}_{2}, the MAP estimate value shifts to $f\sigma_8 \sim 0.35$, and the marginalized distribution becomes bimodal, with two peaks around $f\sigma_8 \sim 0.3$ and $f\sigma_8 \sim 0$. Although this discrepancy may be explained by the prior volume effect, it nevertheless indicates a mild mismatch between the measured ξ^{uu}_{0} and ξ^{uu}_{2}, potentially driven by an unknown effect. Fig. 9 presents the best-fit model and measurements of $\{\xi^{gg}_{0,2}, \xi^{gu}_{1}, \xi^{uu}_{0}\}$, and help clarify the origin of this shift. Although the ξ^{uu}_{2} is not included in the fit, its

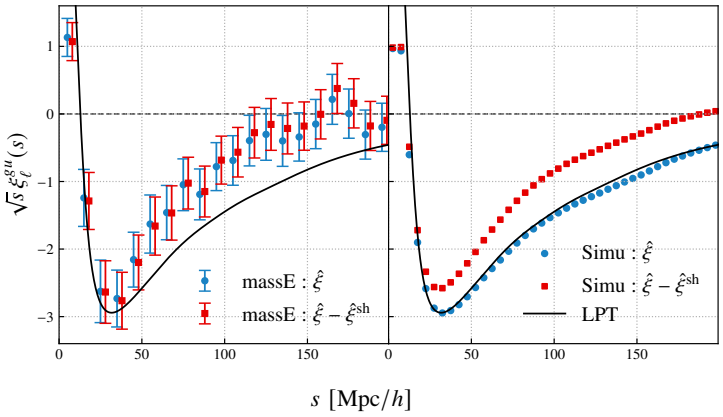


FIG. 11. **Left panel**: Galaxy density-velocity cross-correlation ξ_1^{gu} measured from the `massE` sample before (blue dots) and after (red dots) applying the shuffle-subtraction to null the bulk motion contribution. **Right panel**: Similar measurements from dark matter mocks with the same redshift distribution. As a visual guidance, the black solid lines show the dark matter prediction by one-loop Lagrangian perturbation theory. The small change in the `massE` measurements after shuffle-subtraction suggests that the bulk motions within individual redshift slices are much weaker than expected, owing to either sample variance or systematic bias.

model prediction agrees with the measurement on $s \gtrsim 100\,\mathrm{Mpc}/h$. On scales $s \lesssim 100\,\mathrm{Mpc}/h$, however, the measured quadrupole has a systematically lower amplitude than prediction, thereby favoring an artificially vanishing overall velocity amplitude. As a conservative choice, we therefore exclude ξ_2^{uu} from the subsequent cosmological analysis and adopt the joint fitting to $\{\xi_{0,2}^{gg}, \xi_1^{gu}, \xi_0^{uu}\}$ as our fiducial result.

The joint fitting provides a good description of the data with $\chi^2_{\rm min}/{\rm dof} = 0.84$. Here, ${\rm dof} \equiv N_{\rm d} - N_{\rm p}$, where $N_{\rm d}$ and $N_{\rm p}$ denote the number of data bins and parameters, respectively. The apparent deviation between the measured and modeled ξ_2^{gg} on $s \gtrsim 50\,\mathrm{Mpc}/h$ does not by itself indicate either model failure or systematic bias, because the measurements in different separation bins are strongly correlated through the off-diagonal covariance, as discussed in Ref. [69]. Similarly, different separation bins of ξ_0^{uu} have substantial correlations, so the visually higher data amplitude remains statistically acceptable when the full covariance is considered (see Appendix of Fig. 30). Separate fits to the velocity-only data vector ξ_0^{uu} and density-only data vector $\xi_{0,2}^{gg}$ also provide good descriptions, with $\chi^2_{\rm min}/{\rm dof} = 0.60$ and $\chi^2_{\rm min}/{\rm dof} = 0.98$, respectively. Fig. 10 summarizes the marginalized constraints and the MAP estimates of $f\sigma_8$. The joint and velocity-only analyses yield $f\sigma_8 = 0.40^{+0.06}_{-0.05}$ and $f\sigma_8 = 0.47^{+0.08}_{-0.07}$, with corresponding MAP estimates as $f\sigma_8 = 0.41$ and $f\sigma_8 = 0.52$, respectively. Both results are consistent with the Planck ΛCDM prediction of $f\sigma_8 = 0.45$. By contrast, the density-only analysis gives a substantially lower marginalized constraint, $f\sigma_8 = 0.27^{+0.06}_{-0.10}$, whereas its MAP estimate, $f\sigma_8 \simeq 0.41$, is close to those obtained from the velocity statistics. This discrepancy arises from the strongly non-Gaussian posterior and the associated projection effects, for which the marginalized central value need not coincide with the MAP estimate [62]. The density-only marginalized constraint should therefore be interpreted with caution, while it is tangential to the primary objective of this work.

C. Anomaly of velocity field amplitude: sample variance or systematic bias?

Because the precision of the velocity-field measurements remains limited by the survey volume, most of the cosmological information is effectively encoded in the overall amplitude of the velocity statistics. The resulting constraints shown in Fig. 10 are broadly consistent with one another and with the ΛCDM prediction. Nevertheless, although none of the differences is statistically significant, several mild discrepancies remain in the cosmological interpretation of the peculiar velocity measurements.

- As discussed above, the quadrupole of the velocity auto-correlation has a systematically lower amplitude than the ΛCDM prediction and shifts the inferred $f\sigma_8$ toward lower values.

- The velocity field amplitudes preferred by ξ_0^{uu} and ξ_1^{gu} differ mildly. Specifically, the measured ξ_0^{uu} favors a higher amplitude than the ΛCDM prediction, whereas the measured ξ_1^{gu} favors a lower amplitude. This mismatch could arise from finite volume fluctuations in the density field entering the cross-correlation, although a missing physical or systematic effect cannot be excluded.

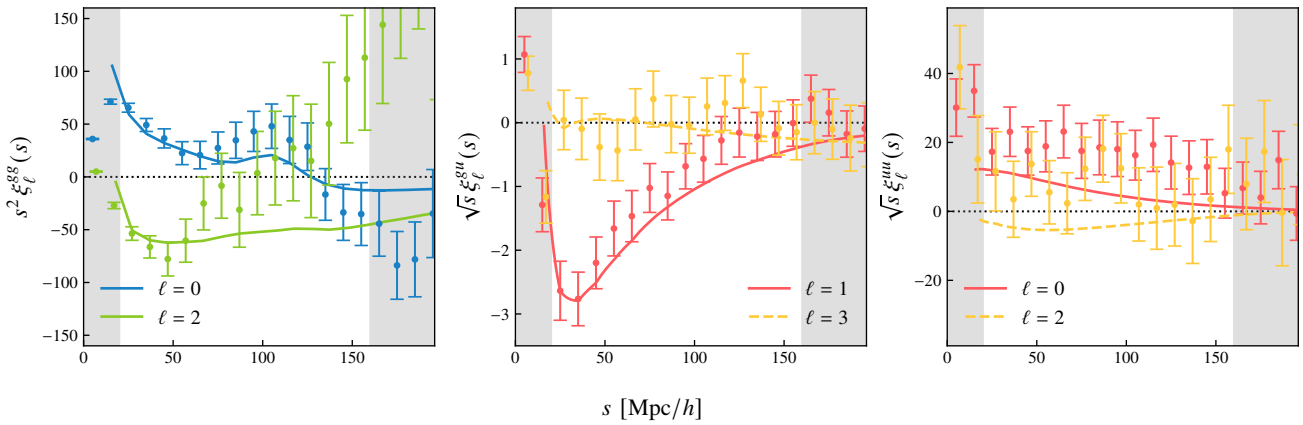


FIG. 12. **Galaxy statistics measured from the `massE` velocity and MGS density catalogs.** Same as Fig. 9, but after applying the shuffle-subtraction procedure to remove the bulk motion from the velocity samples. Although the resulting MAP estimate of $f\sigma_8 = 0.444$ is slightly higher than that in Fig. 9, the predicted amplitude of the velocity statistic is lower and shows better visual agreement with the measurements.

- Although the marginalized constraints from different data combinations show no significant tension in $f\sigma_8$, the full posterior from the joint fit has two peaks, around $f\sigma_8 \sim 0.4$ and $f\sigma_8 \sim 0.45$. This bimodality is statistically allowed by the current uncertainties and parameter degeneracies, and it appears to reflect the different amplitude preferences of ξ_0^{uu} and ξ_1^{gu}.

If they are not resulted from sample variance, one possible explanation for these discrepancies is that the bulk motion $\bar{u}(z)$ averaging over each redshift slice are not fully captured by the current measurements. As discussed in Ref. [28], the limited volume of the SDSS MGS sample prevents a precise determination of the zero-point offset, requiring additional galaxies near the equatorial plane for calibration. This calibration procedure was carefully validated using multiple simulations, and the power spectrum monopole was shown to be robust. However, the quadrupole remains noise-dominated, leaving room for potential shifts. Residual offsets between redshift bins may therefore affect the anisotropic components of the measured velocity field.

To quantify the impact of these mean flows, we subtract the contribution of the mean velocity in each redshift slice by $\hat{\xi} \to \hat{\xi} - \hat{\xi}^{\rm sh}$, where $\hat{\xi}$ denotes the original velocity statistic and $\hat{\xi}^{\rm sh}$ is the mean velocity contribution estimated by shuffling the angle positions of velocity tracers. Fig. 11 presents the resulting density-velocity cross-correlation function before and after this shuffle-subtraction. The change in the observational measurement is much smaller than the prediction of the dark matter mocks. For example, on $s \simeq 100\,{\rm Mpc}/h$, the amplitude of ξ_1^{gu} is almost unchanged by the shuffle-subtraction for the `massE` measurement, while it is reduced by $\sim 50\%$ for the dark matter mock.

As a further test, we jointly fit the galaxy statistics after applying the shuffle-subtraction to both ξ^{gu} and ξ^{uu}. To theoretically model the removal of the bulk motion, one can integrate the velocity statistics over the survey volume, and subtract the resulting mean velocity contribution from the original model prediction. However, this procedure is computationally complex. Instead, we adopt a simplified approach using

$$\xi^{\rm model}(s_i, \mu_i) = \left(\delta^K_{ij} - \mathcal{S}_{ij}\right)\ \xi^{\rm PT}(s_j, \mu_j)\ , \tag{19}$$

where $\xi^{\rm PT}(s_j, \mu_j)$ is the perturbation theory prediction presented in Sec. V A. The shuffle matrix $\mathcal{S}_{ij}$ quantifies how pairs in the j-th (s, μ) bin are mapped into the i-th (s, μ) bin by the shuffling procedure. We estimate $\mathcal{S}_{ij}$ separately for ξ^{gu} and ξ^{uu} by shuffling the angular positions of the velocity tracers 30 times and averaging the results to suppress statistical noise. Fig. 12 presents the joint fit results after the shuffle-subtraction. The marginalized constraint is $f\sigma_8 = 0.39^{+0.06}_{-0.03}$, with a MAP estimate as $f\sigma_8 = 0.44$, consistent with the fiducial constraints without shuffle-subtraction. This consistency is expected because the measurements are noise-dominated, making the overall amplitude constraint insensitive to this modeling variation. However, the measured velocity statistics ξ_1^{gu} and ξ_0^{uu} become more consistent with the ΛCDM prediction after the shuffle-subtraction, particularly ξ_1^{gu}. Furthermore, the posterior in the full parameter space is not bimodal, indicating a unique preferred value of the overall velocity field amplitude.

The suppressed bulk motion may arise either from sample variance, if the bulk motion within the SDSS MGS volume happens to be smaller than the ΛCDM expectation, or from a systematic bias in the measured peculiar velocities, if the bulk motion is not fully captured by calibrations. The limited MGS volume prevents a definitive distinction between

these possibilities, while forthcoming surveys with larger volumes will enable more stringent tests and clarify the origin of the discrepancy. Nevertheless, the illustrative results in Figs. 11 and 12 highlight the importance of calibrating velocity offsets across redshift slices, because these offsets can substantially affect large-scale velocity statistics.

VI. DISCUSSION AND CONCLUSIONS

Galaxy peculiar velocities directly probe the growth of cosmic structure, but their cosmological interpretation requires separating the underlying peculiar motion from measurement systematics. In this work, we perform a detailed investigation of the environmental dependence, underlying angular relations, and overall amplitude of the velocity statistics for peculiar velocities in `massE` catalog. Our main results are summarized as follows.

- The measured velocities depend significantly on local density, with tracers in overdense regions preferentially exhibiting negative velocities. We develop an empirical correction based on the galaxy density templates, that substantially reduces this dependence and largely restores the expected angular antisymmetry of the velocity statistics. A similar trend in the SDSS `FP` velocity catalog suggests that these two distinct distance indicators may share a common source of systematic bias, although its origin remains unclear.

- The characteristic angular relation of the velocity statistics establishes a model-independent diagnostic of peculiar velocity measurements. Even with redshift space distortions and wide angle geometry, the leading anisotropies of the velocity cross- and auto-correlations remain predominantly dipolar and quadrupolar angular structures on large scales, respectively. The dipolar dependence of the density-velocity cross-correlation is detected with high significance for the `massE` catalog on $s \gtrsim 20\,\mathrm{Mpc}/h$, clearly demonstrating that the `massE` peculiar velocity measurements trace the underlying cosmic flow.

- We present the first cosmological analysis combining the full-shape galaxy density and velocity statistics within the Lagrangian bias framework. We adopt the well-established one-loop Lagrangian formulas for the galaxy density statistics, and extend the model to velocity auto-correlations. Simulations spanning different cosmologies and galaxy populations show that the model recovers unbiased structure growth rates. Applied to the `massE` velocity samples, it constrains the structure growth rate to $\sim 15\%$ precision using velocity-only statistics, and $\sim 10\%$ precision when combined with galaxy density, with both results consistent with the Planck ΛCDM prediction.

- Several mild discrepancies remain in the amplitudes of velocity statistics. The suppressed velocity quadrupole, which we exclude from the fiducial cosmological analysis, potentially hints at an unrecognized systematic effect. Moreover, the velocity cross-correlation and auto-correlation prefer different overall amplitudes. Imperfect calibration of bulk motion within redshift slices may partially explain this difference, while leaving the growth rate constraint consistent with the fiducial result. However, current data cannot determine whether the discrepancies arise from sample variance or systematic bias.

Although demonstrated using the `massE` velocity catalog, the methods developed here apply more broadly to peculiar velocity cosmology. The angular signatures follow from rotational symmetry and the vector nature of peculiar motion, providing a model-independent diagnostic for future velocity measurement. The shuffle-subtraction procedure offers a direct but powerful statistical tool to diagnose the bulk motion within a finite volume, by testing the overall amplitude variations. Furthermore, these results not only addresses the challenges in the `massE` peculiar velocity measurements, but also highlight the importance of identifying and controlling systematics for other peculiar velocity surveys. Only by carefully controlling these systematics can peculiar velocities serve as a precision cosmological probe beyond constraints on the overall velocity field amplitude.

Recent progress with DESI further demonstrates the growing potential of peculiar velocity cosmology. Unlike the SDSS MGS sample, of which the number density decreases rapidly at $z \gtrsim 0.1$, the completed DESI survey provides uniform and dense Bright Galaxy Survey (BGS) samples extending to redshift $z \sim 0.4$. This increasing size of BGS samples already enabled large Fundamental Plane and Tully-Fisher samples, and improved the growth measurements at $z < 0.1$ using complementary analysis methods [30–33]. The `massE` relation provides an additional route to enlarge the velocity samples and extend their redshift coverage, as we have demonstrated above. The forthcoming `massE` samples will not only allow tighter measurements of velocity field amplitude, but also provide a more stringent test of potential systematics residuals. Combined with galaxy clustering and weak gravitational lensing, these larger and cleaner peculiar velocities will open new avenues for probing the dark sector and gravity. For example, reconstructing the Newtonian potential across cosmic time, testing the consistency between matter motions and the Weyl potential, and potentially characterizing gravitational screening mechanisms on small scales if environmental systematics in the velocity measurements could be fully understood.

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The authors conducted all conceptualization, data analysis, and interpretation. AI tools were use solely for coding assistance and language polishing. The authors take full responsibility for all results presented in this manuscript.

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Appendix A: Survey sample, two-point statistic and covariance matrix

For the galaxy density catalog, we denote the data and corresponding random samples by D and R, respectively. For the velocity catalog, we denote the velocity components and the position components as V and G, respectively. The galaxy velocity auto-correlation, density-velocity cross-correlation, and density auto-correlation are estimated as

$$\hat{\xi}^{uu}(s,\mu) = \frac{\mathrm{VV}}{\mathrm{GG}} \,, \tag{A1}$$

$$\hat{\xi}^{gu}(s,\mu) = \frac{\mathrm{DV}-\mathrm{RV}}{\mathrm{DG}} \,, \tag{A2}$$

$$\hat{\xi}^{gg}(s,\mu) = \frac{\mathrm{DD}-2\mathrm{DR}+\mathrm{RR}}{\mathrm{RR}} \,, \tag{A3}$$

where s is the pair separation and μ is the cosine between the pair and line-of-sight direction. The normalized pair counts are

$$\mathrm{VV} \equiv \frac{\sum_{(i,j)} w_i^v w_j^v u_i u_j}{N_v^2} \,, \quad \mathrm{GG} \equiv \frac{\sum_{(i,j)} w_i^v w_j^v}{N_v^2} \,, \tag{A4}$$

$$\mathrm{DV} \equiv \frac{\sum_{(i,j)} w_i^g w_j^v u_j}{N_g N_v} \,, \quad \mathrm{DG} \equiv \frac{\sum_{(i,j)} w_i^g w_j^v}{N_g N_v} \,, \quad \mathrm{RV} \equiv \frac{\sum_{(i,j)} w_i^r w_j^v u_j}{N_r N_v} \,, \tag{A5}$$

$$\mathrm{DD} \equiv \frac{\sum_{(i,j)} w_i^g w_j^g}{N_g^2} \,, \quad \mathrm{DR} \equiv \frac{\sum_{(i,j)} w_i^g w_j^r}{N_g N_r} \,, \quad \mathrm{RR} \equiv \frac{\sum_{(i,j)} w_i^r w_j^r}{N_r^2} \,. \tag{A6}$$

Here, $N_g \equiv \sum_i w_i^g$, $N_r \equiv \sum_i w_i^r$ and $N_v \equiv \sum_i w_i^v$ are the summations of object weights in the density catalog, random catalog and velocity catalog, respectively, where w_i^g, w_i^r and w_i^v are the corresponding weights for i-th object. All normalization factors entering the velocity statistics are pair counts between the data catalogs, since there is no random sample constructed for the velocity sample.

After ensemble averaging, the leading-order expectations of three estimators Eqs. (A4, A5 & A6) are

$$\xi^{uu}(s,\mu) \equiv \langle \hat{\xi}^{uu} \rangle \simeq \frac{\langle [1+\delta_v(\mathbf{s}_1)][1+\delta_v(\mathbf{s}_2)] u(\mathbf{s}_1) u(\mathbf{s}_2) \rangle}{1+\langle \delta_v(\mathbf{s}_1)\delta_v(\mathbf{s}_2)\rangle} \tag{A7}$$

$$\xi^{gu}(s,\mu) \equiv \langle \hat{\xi}^{gu} \rangle \simeq \frac{\langle [1+\delta_g(\mathbf{s}_1)][1+\delta_v(\mathbf{s}_2)] u(\mathbf{s}_2) \rangle}{1+\langle \delta_g(\mathbf{s}_1)\delta_v(\mathbf{s}_2)\rangle} \tag{A8}$$

$$\xi^{gg}(s,\mu) \equiv \langle \hat{\xi}^{gg} \rangle \simeq \langle \delta_g(\mathbf{s}_1)\delta_g(\mathbf{s}_2)\rangle \tag{A9}$$

where $\mathbf{s} = \mathbf{s}_2 - \mathbf{s}_1$. Here, δ_g, δ_v and u denote density tracer overdensity, velocity tracer overdensity and line-of-sight peculiar velocity, respectively. The velocity statistics ξ^{uu} and ξ^{gu} are weighted by the tracer number density, and this density-weighting effect becomes significant only on small scales $s \lesssim 10\,\mathrm{Mpc}/h$. On larger scales, the results are reduced to $\xi^{uu} \simeq \langle u_1 u_2 \rangle$ and $\xi^{gu} \simeq \langle \delta_1 u_2 \rangle$.

The pair counts of Eqs. (A4, A5 & A6) and the corresponding two-point correlation functions are computed using the publicly available `Corrfunc` package [76], which performs pair counting in both simulation boxes and realistic curved-sky geometries [77, 78]. To cover the full angular range $-1 \le \mu \le 1$, we use the modified version forked by the DESI Collaboration [79]. Throughout this work, we adopt the midpoint convention to define the line-of-sight direction, i.e., $\hat{n} = (\mathbf{s}_1 + \mathbf{s}_2)/|\mathbf{s}_1 + \mathbf{s}_2|$, where $\mathbf{s}_1$ and $\mathbf{s}_2$ are position vectors of two endpoints.

Pair counts in the data catalogs are weighted by the product of the systematic and Feldman-Kaiser-Peacock (FKP) weights, $w_i \equiv w_i^{\mathrm{sys}} w_i^{\mathrm{FKP}}$, denoted as w^g for density and w^v for velocity catalogs. The systematic weights are provided with the catalogs, while the FKP weights are additionally obtained by $w_i^{\mathrm{FKP}} = \left(\sigma_i^2 + \bar{n} P_0\right)^{-1}$ [3]. Here, $\sigma_i^2 = 1$ for density catalog and $\sigma_i^2(z) = \langle \hat{v}^2 \rangle$ for velocity catalog, where $\hat{v}$ is the measured velocity in unit of km/s. We adopt $P_0 = 10^5\,[\mathrm{Mpc}/h]^3$ and $P_0 = 10^9\,(\mathrm{km/s})^2[\mathrm{Mpc}/h]^3$ for the density and velocity samples, respectively. For the random catalog associated with the density sample, we draw redshifts from the redshift distribution of density sample and compute the corresponding FKP weight for each object. The left panel of Fig. 13 shows the effective number density $\phi(\chi)$ of the `massE` peculiar velocity catalog and the SDSS MGS density catalog. The density sample is signal-dominated, with $\sigma_i^2 \ll \bar{n} P_0$, so $w_i^{\mathrm{FKP}} \simeq \bar{n}^{-1} P_0^{-1}$ and its effective weighting distribution is nearly constant across the redshift range. By contrast, the velocity sample is noise-dominated, giving $w_i^{\mathrm{FKP}} \simeq \sigma_i^{-2}$, equivalent to inverse-variance weighting. Its effective number density therefore decreases with redshift as the measurement uncertainty σ_i increases. These results are insensitive to the choice of P_0.

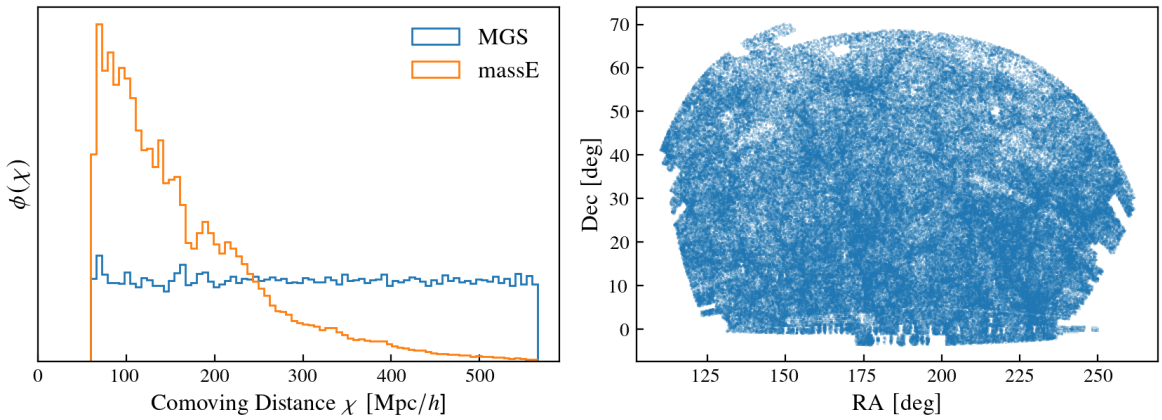


FIG. 13. **Left panel**: Radial distribution of the effective number density for the SDSS MGS density catalog (blue) and the massE peculiar velocity catalog (orange). The effective number density is defined as $\phi \propto \sum_i w_i/\chi_i^2$, where the sample weight w_i includes both systematic and FKP weights. **Right panel**: Sky coverage of the MGS galaxies.

We estimate the covariance matrix of the two-point correlation functions using jackknife resampling, because the measurement uncertainty is dominated by observational noise that is not adequately captured by analytical covariance models. We divide the survey footprint into $N_{\rm JK}$ regions, each containing approximately the same number of galaxies from the density catalog. All galaxies and random points within a given region are assigned to the corresponding jackknife subsample. For each jackknife realization, we exclude one region from both the data and random catalogs and recalculate the correlation function using the remaining regions. Repeating this procedure for all $N_{\rm JK}$ regions yields a set of leave-one-region-out measurements, $\xi_i^{(k)}$, where i labels the correlation function bin and k labels the omitted region. The jackknife covariance matrix is then estimated as

$$\mathbf{Cov}[\xi_i, \xi_j] = \frac{N_{\rm JK}-1}{N_{\rm JK}} \sum_{k=1}^{N_{\rm JK}} \left[\xi_i^{(k)} - \bar{\xi}_i\right]\left[\xi_j^{(k)} - \bar{\xi}_j\right] \quad ,$$

where $\bar{\xi}_i = \frac{1}{N_{\rm JK}} \sum_{k=1}^{N_{\rm JK}} \xi_i^{(k)}$ is the mean over all jackknife realizations. Fig. 30 shows the correlation matrix used in cosmological inference for the fiducial choice of $N_{\rm JK} = 400$. We apply the Hartlap factor attempting to account for the finite number of jackknife realizations when estimating the inverse covariance matrix [80]

$$\mathbf{Cov}^{-1} = \frac{N_{\rm JK} - N_{\rm d} - 2}{N_{\rm JK} - 1} [\mathbf{Cov}]^{-1} \quad ,$$

where $N_{\rm d}$ is the number of correlation function bins. The effects of this and other finite-sample corrections are subdominant because $N_{\rm JK} \gg N_{\rm d}$. Instead, in the cosmological inference, the dominant uncertainty in the covariance matrix arises from the choice of the number of jackknife regions, as discussed below. For applications that require only statistical uncertainties for qualitative comparisons, we instead estimate the covariance directly from $N_{\rm sub} = 36$ individual subsamples, avoiding the substantial computational cost of full jackknife resampling. Galaxy mocks that jointly reproduce the observed density and velocity fields are ultimately required for robust covariance estimation and cosmological inference. However, constructing such mocks is beyond the scope of current analysis, and is left for future work with forthcoming larger samples.

Appendix B: A method to alleviate the density-dependent systematics

In this appendix, we describe details of the empirical correction for the density-dependent velocity residuals identified in Sec. III. For the peculiar velocity field, conditioning on a local scalar density does not select a preferred direction, so its ensemble mean is expected to vanish. We use this expectation as a null condition to construct a template that suppresses the measured conditional mean velocities across density components on different scales, thereby mitigating the density-dependent systematics.

As demonstrated in Sec. III, we firstly decompose the galaxy density field $\delta_g(\mathbf{s})$ into a set of filtered fields, spanning a range of spatial scales, by filtering the galaxy overdensity fields with Gaussian band-pass filters

$$\delta_g^\alpha(\mathbf{s}) = W_\alpha(\mathbf{s}) * \delta_g(\mathbf{s}) \ , \quad W_\alpha(k) = \exp\left[-\frac{(\ln k - \ln k_\alpha)^2}{2\sigma_W^2}\right] \ .$$

where $\alpha = 1, \ldots, N_\delta$ labels the density filters, and $\sigma_W \equiv 3.6/N_\delta$ is the bandwidth, chosen such that $\sum_{\alpha=1}^{N_\delta} W_\alpha(k)$ remains approximately constant over the scales in analysis. As shown in Fig. 14, different filters produce the density fluctuations at different spatial scales, so that we can capture the scale variation of the density-dependent systematics. For each filtered field, we divide the velocity tracers into bins of $\delta_g^\alpha(\mathbf{s}_i)$, where $\mathbf{s}_i$ denote the positions of the velocity tracers. Let $\mathcal{B}_a^\alpha$ denote the set of velocity tracers in the a-th bin of the α-th filtered density field, we define the conditional mean as

$$\Delta u_a^\alpha \equiv \frac{\sum_{i\in\mathcal{B}_a^\alpha} w_i^v \, u(\mathbf{s}_i)}{\sum_{i\in\mathcal{B}_a^\alpha} w_i^v} \tag{B1}$$

where w_i^v is the weight of i-th velocity tracer, as defined in Appendix. A. Assigning the corresponding bin average to each tracer, labeled by its position $\mathbf{s}_i$, we can define the α-th template

$$\Delta u^\alpha(\mathbf{s}_i) \equiv \Delta u_a^\alpha \ , \qquad i \in \mathcal{B}_a^\alpha . \tag{B2}$$

Here, Δu_a^α is a binned statistic, while $\Delta u^\alpha(\mathbf{s}_i)$ is a catalog sampled by the local density component. We model the density-dependent contamination as the linear combination of N_δ templates

$$\Delta\hat{u}(\mathbf{s}_i) = \sum_{\alpha=1}^{N_\delta} \mathcal{W}_\alpha \, \Delta u^\alpha(\mathbf{s}_i) \ , \tag{B3}$$

and the peculiar velocity is therefore corrected as

$$u_{\rm c}(\mathbf{s}_i) = u(\mathbf{s}_i) - \Delta\hat{u}(\mathbf{s}_i) \ . \tag{B4}$$

The coefficients $\mathcal{W}_\alpha$ are chosen such that Eq. (B3) serves an effective template to suppress the density-dependent systematics, which is realized by requiring the conditional averaging of the corrected velocities Eq. (B4) to vanish across all filtered density fields,

To solve for the coefficients $\mathcal{W}_\alpha$, we define the conditional mean of the corrected velocities with respect to the filtered density δ_g^β, similar to Eq. (B1), as

$$\varepsilon_a^\beta \equiv \frac{\sum_{i\in\mathcal{B}_a^\beta} w_i^v \, u_{\rm c}(\mathbf{s}_i)}{\sum_{i\in\mathcal{B}_a^\beta} w_i^v} = \Delta u_a^\beta - \sum_{\alpha=1}^{N_\delta} \mathcal{W}_\alpha \Delta u_a^{\alpha\beta} . \tag{B5}$$

where the conditional mean of the α-th template with respect to the filtered density δ_g^β is defined accordingly as

$$\Delta u_a^{\alpha\beta} \equiv \frac{\sum_{i\in\mathcal{B}_a^\beta} w_i^v \, \Delta u^\alpha(\mathbf{s}_i)}{\sum_{i\in\mathcal{B}_a^\beta} w_i^v} \tag{B6}$$

The expectation that the conditional averaging of the corrected velocities vanishes imposes the condition $\langle\varepsilon_a^\beta\rangle = 0$, which gives a set of linear equations,

$$\Delta u_a^\beta = \sum_{\alpha=1}^{N_\delta} \mathcal{W}_\alpha \Delta u_a^{\alpha\beta} \quad , \tag{B7}$$

for every a-th bin of every β-th filtered density field. Because both Δu_a^β and $\Delta u_a^{\alpha\beta}$ are directly measured from the velocity data, we can therefore obtain the coefficients $\mathcal{W}_\alpha$ by solving linear equations of Eq. (B7).

The peculiar velocities and the corresponding statistics are noisy, so we need to account for the statistical uncertainties when solving Eq. (B7). As a simplification, we further assume that the fluctuations of ε_a^β are Gaussian distributed. The covariance matrix of ε_a^β is

$$\begin{aligned}\mathbf{Cov}_{\alpha a\beta b}(\mathcal{W}) \equiv \mathbf{Cov}[\varepsilon_a^\alpha, \varepsilon_b^\beta] = \mathbf{Cov}[\Delta u_a^\alpha, \Delta u_b^\beta] + \sum_{\gamma\lambda} \mathcal{W}_\gamma \mathcal{W}_\lambda \mathbf{Cov}[\Delta u_a^{\gamma\alpha}, \Delta u_b^{\lambda\beta}] \\ - \sum_\gamma \mathcal{W}_\gamma \mathbf{Cov}[\Delta u_a^{\gamma\alpha}, \Delta u_b^\beta] - \sum_\lambda \mathcal{W}_\lambda \mathbf{Cov}[\Delta u_a^\alpha, \Delta u_b^{\lambda\beta}] \end{aligned} \tag{B8}$$

Here, on the right side of Eq. (B8), four kinds of covariances of the conditional mean can be also measured from velocity data. The coefficients are therefore obtained by minimizing the negative log-likelihood,

$$-2\ln\mathcal{L}(\mathcal{W}) \;=\; \sum_{\alpha i\beta j} \varepsilon_i^\alpha \, [\mathbf{Cov}]^{-1}_{\alpha i\beta j} \, \varepsilon_j^\beta + \ln\det \mathbf{Cov} + \text{const} \quad . \tag{B9}$$

In practice, we estimate all covariance terms in Eq. B8 using 2400 jackknife realizations. Because the filtered overdensity fields δ_g^α span different ranges and follow different distributions, applying a common uniform binning scheme would produce highly imbalanced numbers of velocity tracers across bins, leading to numerical instability in the covariance matrix. To mitigate this issue, we transform each δ_g^α field to follow a standard Gaussian distribution while preserving the rank order of the original values. We then divide the transformed values into 18 uniformly spaced bins over $[-3, 3]$, excluding values outside this range. This procedure produces a more balanced distribution of velocity tracers across bins, stabilizing both the covariance estimate and the likelihood optimization in Eq. (B9) used to determine the coefficients $\mathcal{W}_\alpha$.

Fig. 15 shows the conditional mean velocities Δu_a^α before correction and the residual means ε_a^α after correction for the fiducial choice $N_\delta = 12$. The correction substantially suppresses the prominent density-dependent trends across the filtered fields, demonstrating that the simple linear templates capture a large fraction of the environmental dependence. Fig. 16 shows the normalized residual covariance matrix of Eq. (B8), evaluated at the optimal coefficients $\mathcal{W}_\alpha$ obtained by minimizing Eq. (B9). To assess sensitivity to the number of density filters, we repeat the analysis for $N_\delta \in \{6, 12, 16, 20, 24, 30\}$. Fig. 17 compares the inferred corrections $\Delta\hat{u}(\mathbf{s}_i)$ obtained with different numbers of filters. Although the scatter shows some dependence on the sampling of density scales, the corrections remain strongly correlated, with not appreciable systematic offsets between the filter choices. This agreement demonstrates that the density-dependence correction is robust against the specific choice of N_δ.

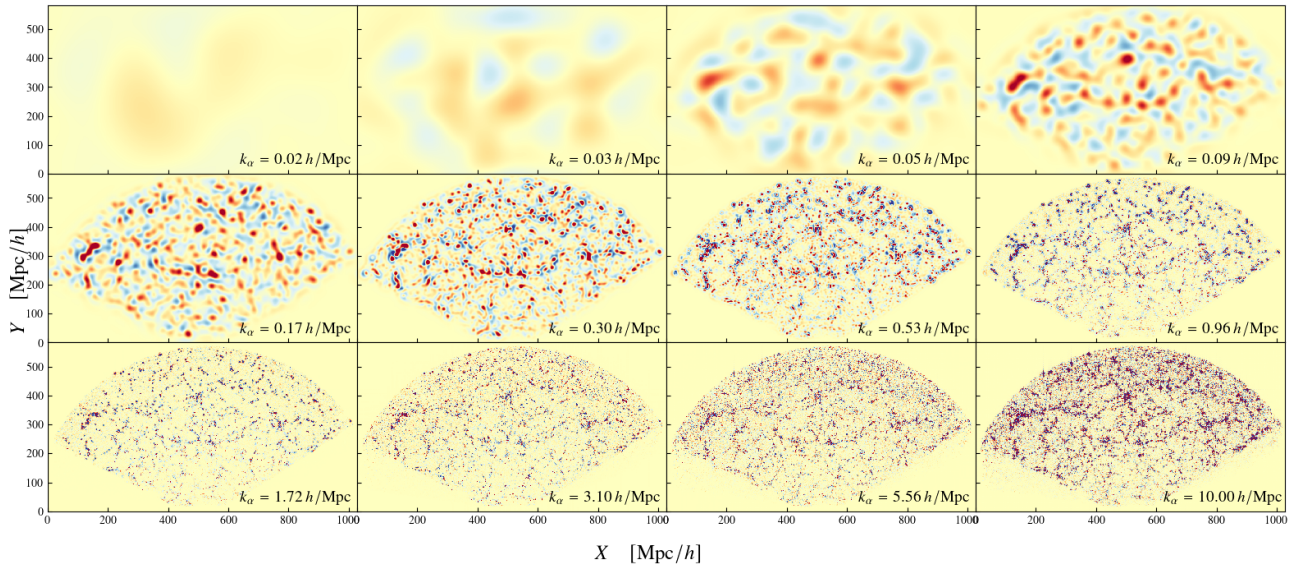


FIG. 14. **Filtered galaxy overdensity fields used to construct the density-dependent velocity templates.** Each panel shows a density component of δ_g^α, obtained by applying the Gaussian band-pass filter $W_\alpha(k) = \exp\left[-\frac{(\ln k - \ln k_\alpha)^2}{2\sigma_W^2}\right]$ in Fourier space. The characteristic wavenumbers are uniformly spaced in $\log_{10}[k_\alpha/(h\,\mathrm{Mpc}^{-1})]$ over $[-1.8, 1]$, in total $N_\delta = 12$ bands with bandwidth $\sigma_W = 0.3$. Panels with increasing k_α illustrate the density fluctuations selected on progressively smaller spatial scales. For visualization, the color scale is normalized independently in each panel.

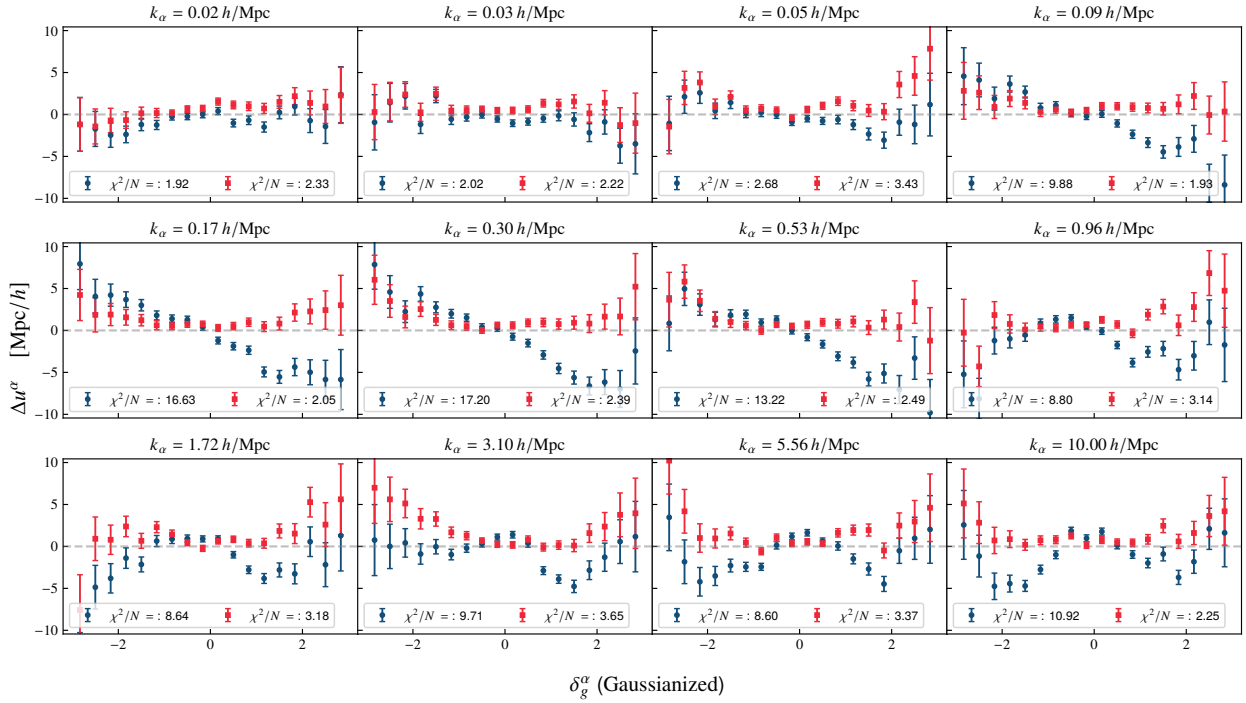


FIG. 15. **Conditional mean velocity Δu^α for each filtered density field δ_g^α** , before and after the density-dependence correction for the fiducial choice $N_\delta = 12$. Each panel corresponds to a different filtered overdensity field, where the sampled $\delta_g^\alpha(\mathbf{s}_i)$ is transformed to follow a Gaussian distribution with zero mean and unit variance. Blue circles show the uncorrected conditional means Δu_a^α in each bin, while red squares show the residuals ε_a^α after subtracting the inferred velocity corrections.

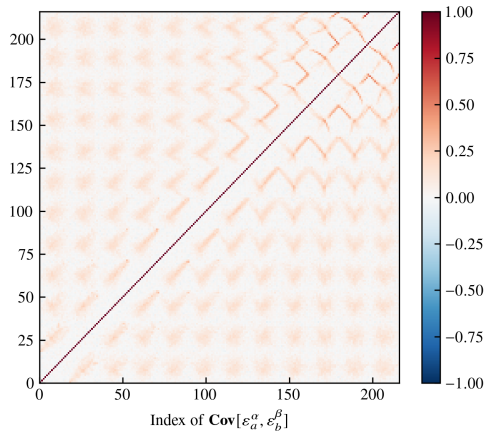


FIG. 16. **Covariance matrix of Eq. (B8) for the binned conditional residuals** ε_a^α entering the likelihood used to determine the correction coefficients $\mathcal{W}_\alpha$. Here, we present the normalized covariance matrix $\mathbf{Cov}[\varepsilon_a^\alpha, \varepsilon_b^\beta]/\sqrt{\sigma^2_{\varepsilon_a^\alpha}\sigma^2_{\varepsilon_b^\beta}}$ for $N_\delta = 12$, estimated from 2400 jackknife resampling times. Both axes index the overdensity bins stacked across all filtered density fields.

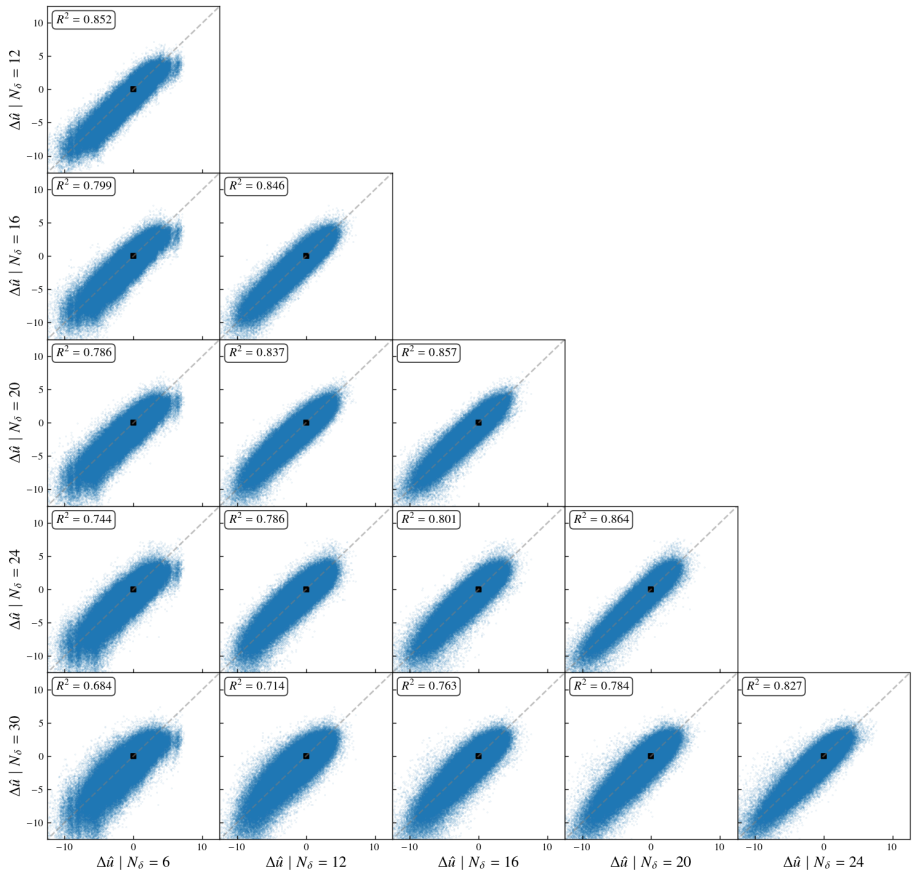


FIG. 17. **Comparison of the inferred velocity corrections $\Delta\hat{u}(\mathbf{s}_i)$ obtained using different numbers of density filters.** Each panel compares the corrections $\Delta\hat{u}(\mathbf{s}_i)$ in Eq. (B3) for individual velocity tracers between two choices of $N_\delta \in \{6, 12, 16, 20, 24, 30\}$. Blue points represent individual tracers, and the dashed diagonal line denotes equality between the two estimates. The cross correlation coefficient R^2 between two datasets is also reported for each comparison. Although the scatter indicates some sensitivity to the sampling of density scales, the corrections remain strongly correlated, without any appreciable systematic offset between the different filter choices.

Appendix C: Underlying angular relations of the peculiar velocity statistics

In the main text, we assess the impact of both redshift space distortions and wide angle effects on the characteristic angular relations of the peculiar velocity statistics using the dark matter snapshot from the CosmicGrowth-6620 simulation [53] at $z = 0.12$. The simulation contains 3072^3 particles in a box of size $L = 1200\,h^{-1}$Mpc, and adopts a flat WMAP cosmology [81], with $\Omega_b = 0.0445$, $\Omega_c = 0.2235$, $h = 0.71$, and $\sigma_8 = 0.83$. In this appendix, to isolate the individual effects, we consider four different configurations for the correlation function measurement, as illustrated in Fig. 18. We test (a) real-space measurements with the flat sky approximation, (b) redshift-space measurements with the flat sky approximation, (c) real-space measurements in curved sky, and (d) redshift-space measurements in curved sky. The last configuration corresponds to the mock results presented in Fig. 5, including both effects. For the curved sky measurements, we place an observer at a fixed position, and select particles within range $0.02 < z < 0.2$. The sky coverage is $f_{\rm sky} = 1/8$, approximately same as the survey samples. With treating the particles with unit weights, we approximate the density tracers as having a uniform comoving number density, and downsample the velocity tracers to match the observed effective redshift distribution, shown in Fig. 13. For each configuration test, we repeat the measurements using 64 different observer positions and average the results to reduce sample variance.

Fig. 19 presents the dark matter density-velocity correlation of $\xi^{gu}(s,\mu)$ for these four configurations. As theoretical expectations, the real-space measurements in flat sky accurately recover the relation $\xi^{gu}(s,\mu) \propto \mu$. Redshift space distortions generate substantial departures at $s \lesssim 20\,{\rm Mpc}/h$, particularly for pairs close to the line-of-sight, while the relation remains a good approximation at $s \gtrsim 20\,{\rm Mpc}/h$. Wide angle effects are mild and become visible mainly at $s \gtrsim 100\,{\rm Mpc}/h$. Fig. 20 shows the similar results for $\xi^{uu}(s,\mu)$. The real-space results in flat sky recover $\xi^{uu}(s,\mu) - \xi^{uu}_0(s) \propto \mu^2 - 1/3$. Wide angle effects again introduce only small corrections. The nonlinear redshift space distortions produce a U-shaped feature toward small scales, becoming pronounced at $s \lesssim 20\,{\rm Mpc}/h$. On scales $s \gtrsim 20\,{\rm Mpc}/h$, the quadrupolar approximation of ξ^{uu} broadly remains for all configurations. Therefore, the dominant contaminations are redshift space distortions, particularly their nonlinear effects, while the wide angle effects and their interplay with redshift space distortions remain well below the observational uncertainties.

We further test the impact of nonlinear redshift space distortions for various density tracers, using galaxy samples selected from the `L1_m9` simulation in the FLAMINGO suite, with simulation details given in Appendix. D 4. Figs. 21 & 22 show the residual correlation functions after subtracting lower-order multipoles of the same parity, as defined in Eqs. (11 & 12). For simplicity, we perform these tests in the flat sky approximation, because the preceding analysis shows that redshift space distortions dominate the departures. On scales $s \gtrsim 20\,{\rm Mpc}/h$, the leading dipolar and quadrupolar structures remain consistent across the samples, and agree with the dark matter results in Fig. 5. On scales $s \simeq 5\,{\rm Mpc}/h$, however, their behavior differs substantially because of different nonlinear clustering among populations. The consistency between the dark matter and galaxy results reflects that the angular relations of the velocity statistics arise from rotational symmetry, rather than determined by the specific clustering properties, and is therefore insensitive to tracer selection. Altogether, on scales $s \gtrsim 20\,{\rm Mpc}/h$, the characteristic angular relations are robust to redshift space distortions, wide angle effects, and tracer selection on large scales, supporting its use as a diagnostic of peculiar velocity measurements.

In Fig. 23, 24 & 25, we repeat the `massE` sample analysis using the SDSS `FP` peculiar velocity sample from Ref. [27]. We find similar density dependent systematics, which are substantially reduced by our empirical correction. The characteristic angular relations are also recovered, although at lower significance because of the smaller sample size and survey volume.

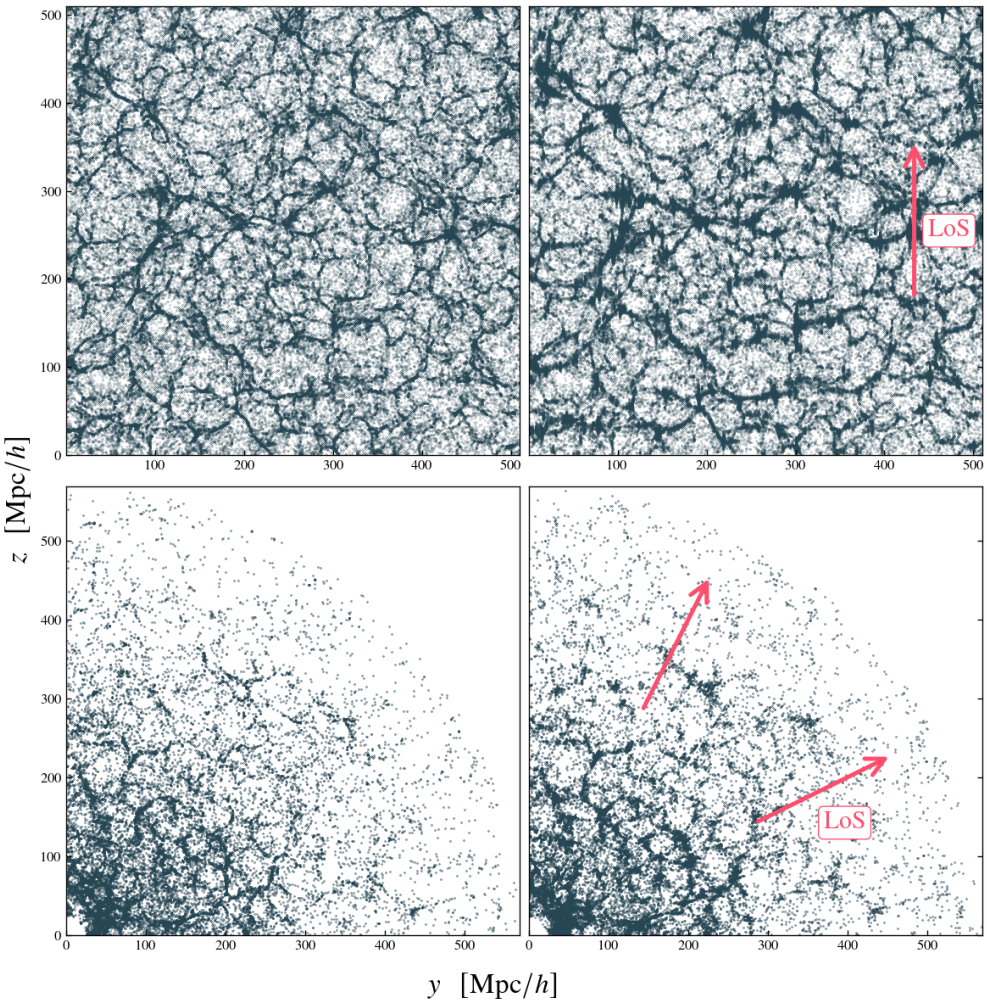


FIG. 18. **Illustration of four different configurations for correlation function measurements in the simulation validation. Top left:** real-space measurement with flat sky approximation, where the line-of-sight direction is fixed along the z-axis. **Top right:** redshift-space measurement with flat sky approximation. **Bottom left:** real-space measurement in curved sky within $0.02 < z < 0.2$, where the line-of-sight direction varies across the observer's sky. The radial distribution is downsampled to match the effective redshift distribution of the velocity sample. **Bottom right:** redshift-space measurement in curved sky within $0.02 < z < 0.2$. All snapshots shown here adopt the projected depth $\Delta h = 6\,\mathrm{Mpc}/h$.

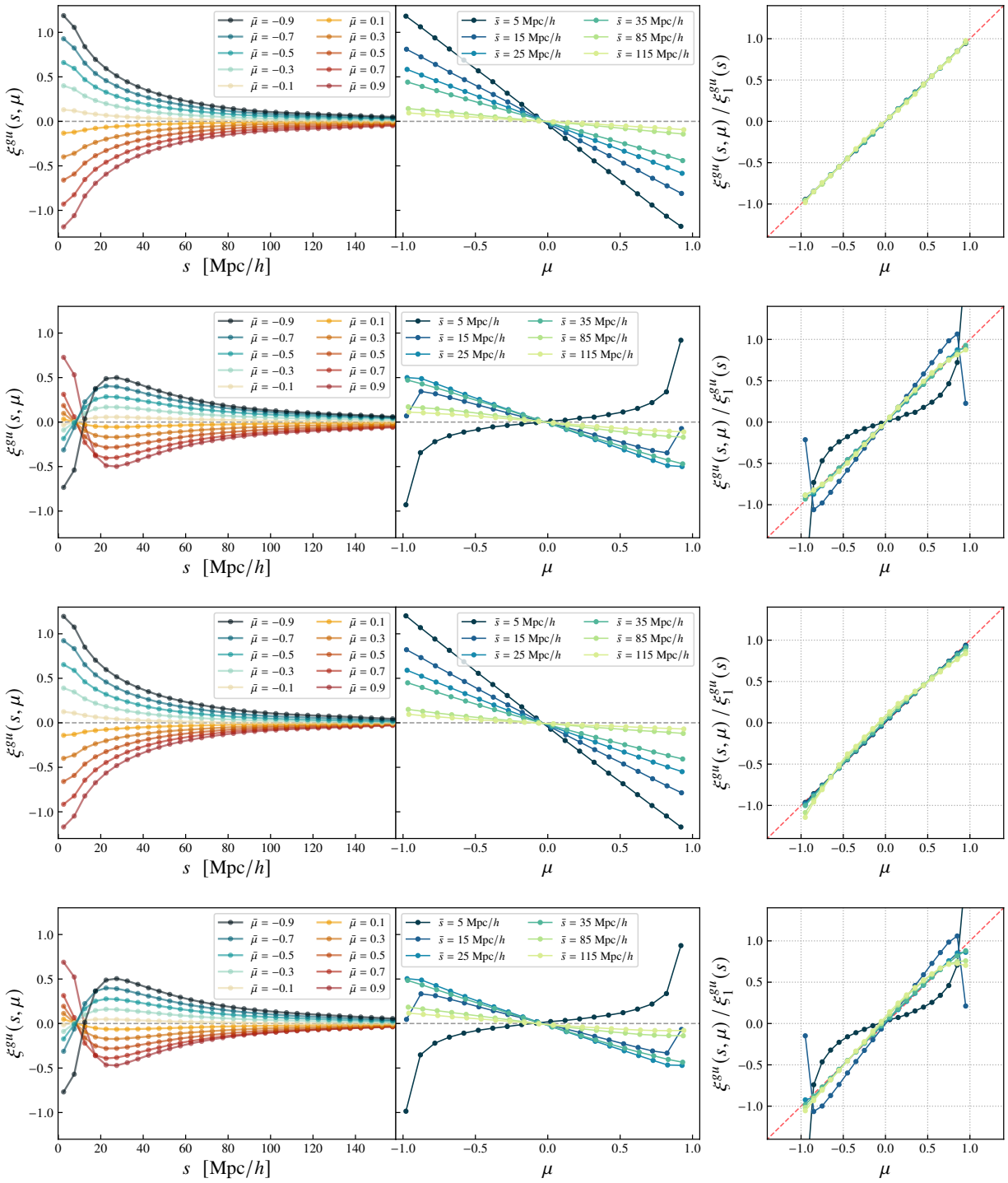


FIG. 19. **Simulation measurements of density-velocity cross-correlation function in four different configurations**, as illustrated in four panels of Fig. 18. The left column presents $\xi^{gu}(s,\mu)$ for different μ bins with width $\Delta\mu = 0.2$. The middle column presents the angular dependence for different s bins with width $\Delta s = 10\,\mathrm{Mpc}/h$. The right column presents the angular dependence by eliminating the overall amplitude at the given s-bin. **First row**: real-space measurement with flat sky approximation. The angular dependence $\xi^{gu}(s,\mu) \propto \mu$ is strictly recovered in the absence of observational effects. **Second row**: redshift-space measurement with flat sky approximation. The redshift space distortions significantly modify the shape of $\xi^{gu}(s,\mu)$ on small scales. However, the angular dependence still maintains for the large separation $s \gtrsim 20\,\mathrm{Mpc}/h$. **Third row**: real-space measurement in curved sky. The wide angle effects only have mild impact on the angular dependence for $s \gtrsim 100\,\mathrm{Mpc}/h$. **Fourth row**: redshift-space measurement in curved sky. Although both the redshift space distortions and wide angle effects contribute to the angular dependence, the relation $\xi^{gu}(s,\mu) \propto \mu$ still retain for $s \gtrsim 20\,\mathrm{Mpc}/h$.

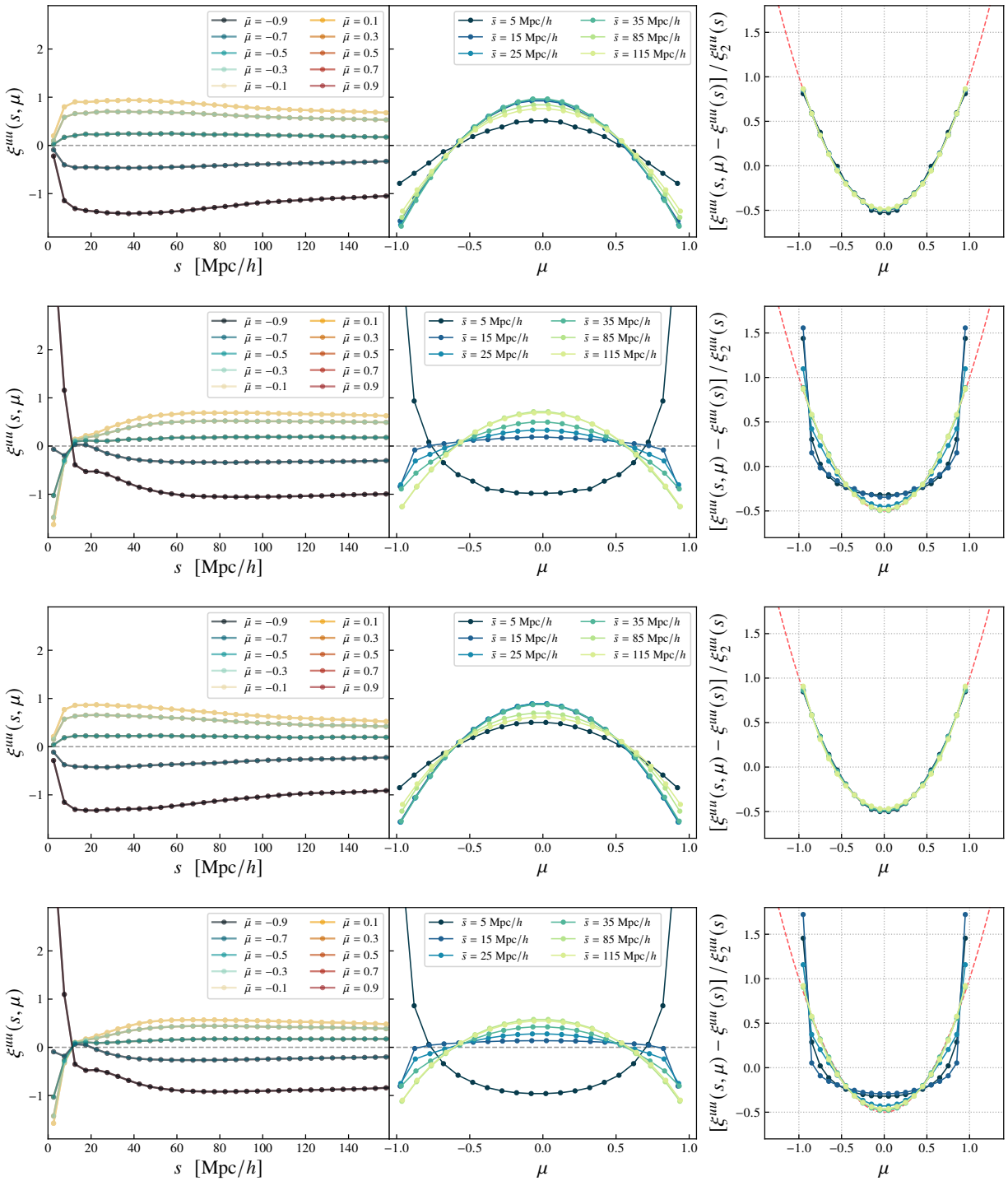


FIG. 20. **Simulation measurements of velocity auto-correlation function in four different configurations**, similar to Fig. 19.

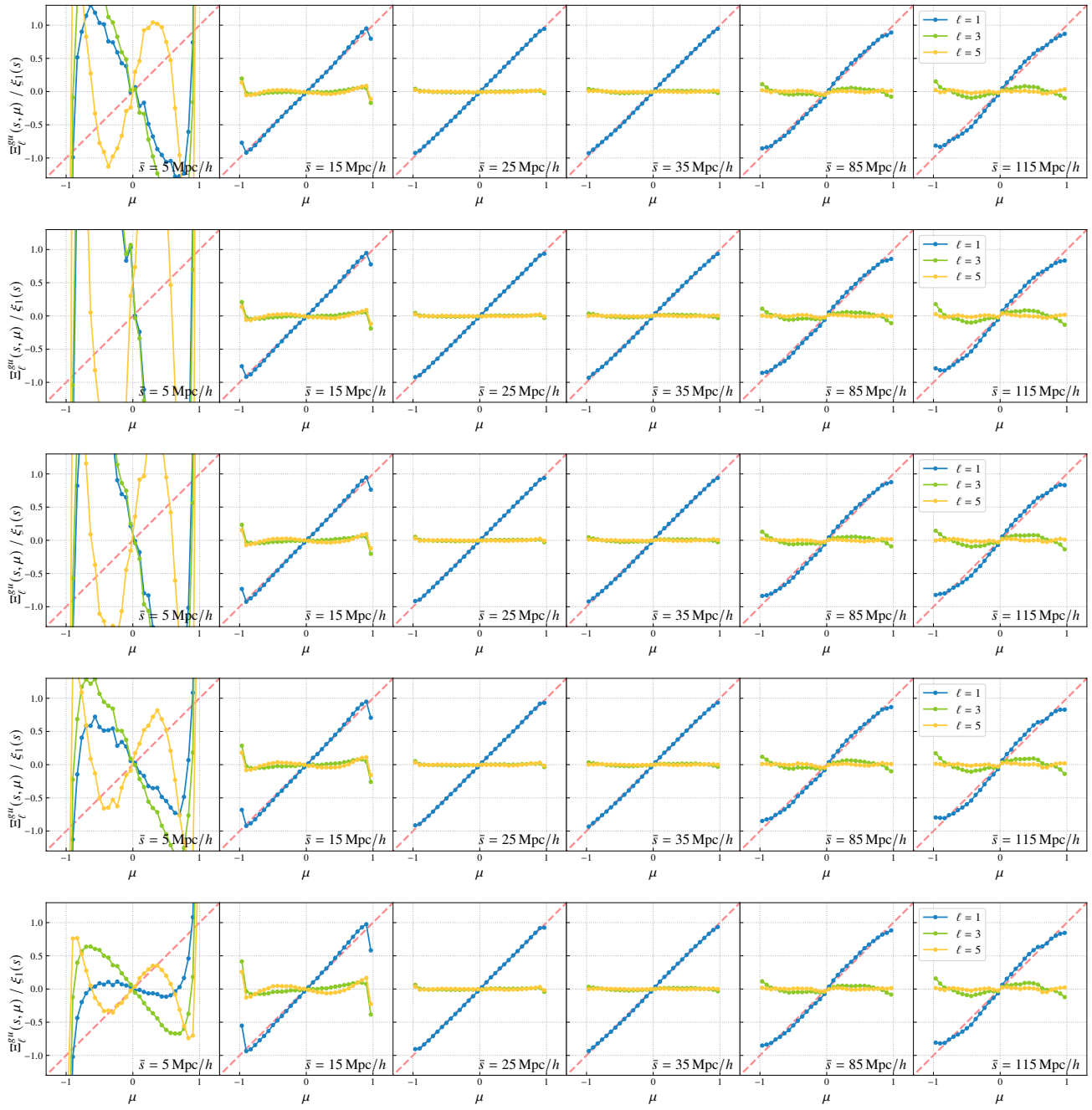


FIG. 21. **Validation of the angular relations $\xi^{gu}(s,\mu) \propto \mu$ using the FLAMINGO `L1_m9` galaxy samples**, similar to Fig. 5 but with only redshift space distortions. Rows from top to bottom correspond to the five stellar-mass-selected subsamples with decreasing stellar mass and clustering bias, while panels from left to right show measurements in different separation bins $\bar{s}$ with width $\Delta s = 10\,\mathrm{Mpc}/h$. In each panel, solid lines show the residual correlation functions for ξ^{gu} defined in Eq. (11), and red dashed lines show the theoretical expectations for the leading-order angular structure.

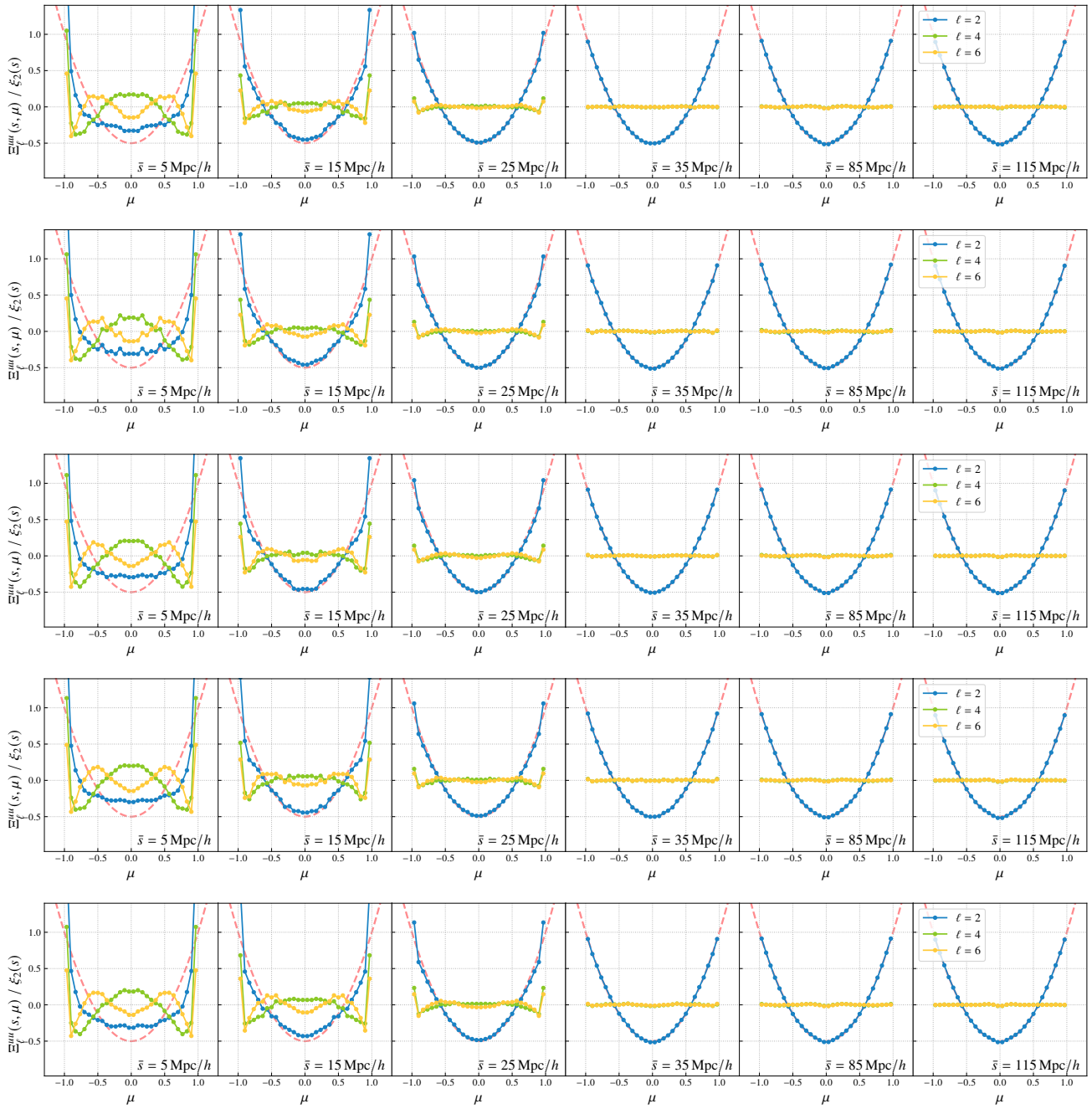


FIG. 22. **Validation of the angular relations $\xi^{uu}(s,\mu) - \xi^{uu}_0(s) \propto \mu^2 - \frac{1}{3}$ using the FLAMINGO `L1_m9` galaxy samples**, similar to Fig. 5 but with only redshift space distortions. Rows from top to bottom correspond to the five stellar-mass-selected subsamples with decreasing stellar mass and clustering bias, while panels from left to right show measurements in different separation bins $\bar{s}$ with width $\Delta s = 10\,\mathrm{Mpc}/h$. In each panel, solid lines show the residual correlation functions for ξ^{uu} defined in Eq. (12), and red dashed lines show the theoretical expectations for the leading-order angular structure.

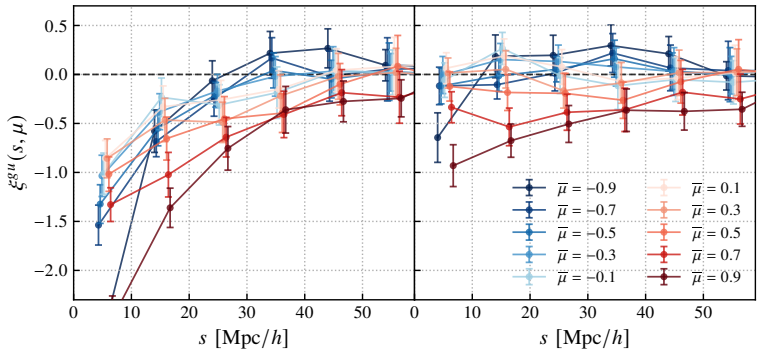


FIG. 23. Similar to Fig. 1, but present the results for **the SDSS FP peculiar velocity sample** from Ref. [27].

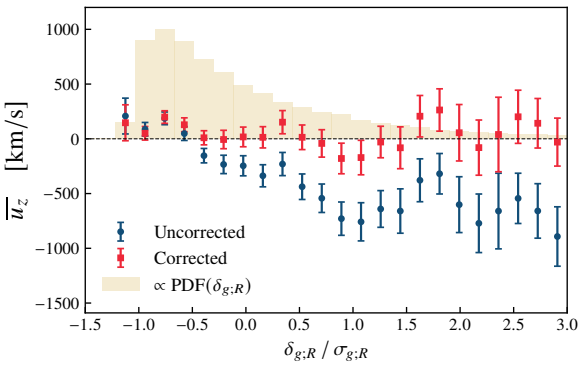


FIG. 24. Similar to Fig. 2, but present the results for **the SDSS FP peculiar velocity sample** from Ref. [27].

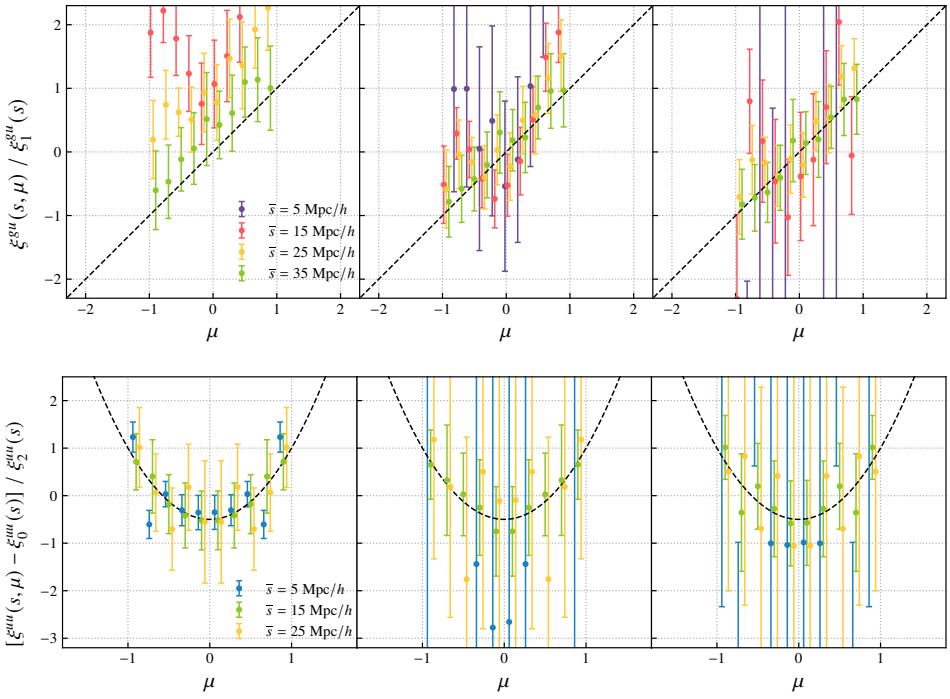


FIG. 25. Similar to Fig. 6, but present the results for **the SDSS FP peculiar velocity sample** from Ref. [27]. Panels from left to right present the multipoles obtained from a single FP fit, from the same catalog after our density-dependence correction, and from separate FP fits in group-richness bins.

Appendix D: Modeling the velocity and density statistics in redshift space

We present details of the theoretical model applied to the cosmological analysis, organized as follows. In Sec. D 1, we provide the general treatment for modeling the redshift-space statistics of the galaxy density and peculiar velocity fields. In Sec. D 2, we introduce the Lagrangian bias expansion, present explicit expressions for the power spectra, and specify the associated model parameters. In Sec. D 3, we describe the density-normalization correction required for the velocity statistics. Finally, in Sec. D 4, we validate the theoretical model using the FLAMINGO simulations and assess its ability to recover the underlying structure growth rate across multiple cosmologies and galaxy samples.

1. Redshift-space statistics

We adopt the well established Lagrangian bias expansion theory to model the redshift-space galaxy statistics [47, 54–66], therefore extracting the underlying cosmological information. We first summarize the relevant formulas, and then extend them to the velocity auto-correlation.

In redshift space, the observed galaxy position is distorted by the peculiar motion as $\mathbf{s} = \mathbf{x} + u\,\hat{n}$, where $\mathbf{x}$ is its real-space postion, $\hat{n}$ is the line-of-sight direction, and $u = \mathcal{H}^{-1}\mathbf{v}\cdot\hat{n}$ is the line-of-sight peculiar velocity in units of conformal Hubble parameter. In the Lagrangian description, the real-space position is related to the initial coordinate $\mathbf{q}$ through $\mathbf{x} = \mathbf{q} + \mathbf{\Psi}$, where $\mathbf{\Psi}$ is the Lagrangian displacement field. The redshift-space mapping can therefore be written as $\mathbf{s} = \mathbf{q} + \mathbf{\Psi}^s(\mathbf{q})$, where the redshift-space displacement is accordingly $\mathbf{\Psi}^s \equiv \mathbf{\Psi} + u\,\hat{n}$. Let $\delta_g(\mathbf{s})$ denote the observed galaxy overdensity in redshift space and $\delta_g^L(\mathbf{q})$ the corresponding overdensity in Lagrangian space, the velocity and density two-point statistics can be written in the unified form,

$$P^s_{LL'}(\mathbf{k}) = \int_{\mathbf{s}} \langle\, [1+\delta_g(\mathbf{s}_1)]\,[1+\delta_g(\mathbf{s}_2)]\; u^L(\mathbf{s}_1) u^{L'}(\mathbf{s}_2)\,\rangle\, e^{i\mathbf{k}\cdot\mathbf{s}} \tag{D1}$$

$$= \int_{\mathbf{q}} \langle\, \left[1+\delta_g^L(\mathbf{q}_1)\right]\,\left[1+\delta_g^L(\mathbf{q}_2)\right]\; u^L(\mathbf{q}_1) u^{L'}(\mathbf{q}_2)\, e^{i\mathbf{k}\cdot\Delta\mathbf{\Psi}^s}\,\rangle\, e^{i\mathbf{k}\cdot\mathbf{q}} \tag{D2}$$

where $\mathbf{q} = \mathbf{q}_2 - \mathbf{q}_1$, $\mathbf{s} = \mathbf{s}_2 - \mathbf{s}_1$, and $\Delta\mathbf{\Psi}^s = \mathbf{\Psi}^s_2 - \mathbf{\Psi}^s_1$. For simplicity, we adopt the shorthand notation for integrals $\int_{\mathbf{s}} \equiv \int d^3\mathbf{s}$ in configuration space and $\int_{\mathbf{k}} \equiv \int \frac{d^3\mathbf{k}}{(2\pi)^3}$ in Fourier space. In the second equality, we have transformed the integral from redshift-space coordinate of $\mathbf{s}$ to Lagrangian coordinate of $\mathbf{q}$ by the number conservation of the biased tracers, $1+\delta_g(\mathbf{s}) = \int_{\mathbf{q}} \left[1+\delta_g^L(\mathbf{q})\right]\delta^{\rm D}\left[\mathbf{s}-\mathbf{q}-\mathbf{\Psi}^s(\mathbf{q})\right]$. In the Lagrangian formula of Eq. (D2), the peculiar velocity are treated in the same manner as the displacement $\mathbf{\Psi}$ by $\mathbf{v} = a\dot{\mathbf{\Psi}}$. The time dependence of all fields is suppressed for notational simplicity. Different choices of power index (L, L') yield different density-weighted velocity, e.g.,

- $(L, L') = (0, 0)$: galaxy density auto-correlation $P^s_{00}(\mathbf{k}) = P^s_{gg}(\mathbf{k})$
- $(L, L') = (1, 0)$: galaxy density-velocity cross-correlation $P^s_{10}(\mathbf{k}) = P^s_{gu}(\mathbf{k})$
- $(L, L') = (1, 1)$: galaxy velocity auto-correlation $P^s_{11}(\mathbf{k}) = P^s_{uu}(\mathbf{k})$

These three spectra, $P^s_{gg}(\mathbf{k})$, $P^s_{gu}(\mathbf{k})$, and $P^s_{uu}(\mathbf{k})$, constitute the theoretical observables applied to the cosmological analysis presented in the main text. For the redshift-space galaxy power spectrum P^s_{gg}, we adopt the one-loop Lagrangian perturbation theory developed in Refs. [58–61]. Their theoretical implementation directly evaluates the Lagrangian perturbation theory integrals with efficient algorithms, which resums both the long-wavelength displacements responsible for matter clustering and the long wavelength velocity arising from redshift space distortions. Various tests based on simulations, together with applications to survey data, have demonstrated that this theoretical model achieves sub-percent accuracy [61–65, 82], far exceeding the requirements of our analysis.

Since the velocity is proportional to the linear growth rate, the pairwise velocity statistics in redshift space can be generated by the derivatives of the redshift-space density power spectrum [47, 66],

$$\Xi^{(n)}_s(\mathbf{k}) \equiv \int_{\mathbf{s}} \langle\, [1+\delta_g(\mathbf{s}_1)]\,[1+\delta_g(\mathbf{s}_2)]\; (u_2-u_1)^n\,\rangle\, e^{i\mathbf{k}\cdot\mathbf{s}} \tag{D3}$$

$$= \frac{1}{(ik\mu_{\mathbf{k}})^n}\frac{{\rm d}^n}{{\rm d}\lambda^n} P^s(\mathbf{k};\lambda)\Big|_{\lambda=1} \tag{D4}$$

where the generating function, a generalized density power spectrum, is obtained by a simple replacement $f \to \lambda f$,

$$P^s(\mathbf{k};\lambda) = \int_{\mathbf{x}} \langle\, [1+\delta_g(\mathbf{x}_1)]\,[1+\delta_g(\mathbf{x}_2)]\; e^{i\mathbf{k}\cdot\lambda\Delta\mathbf{u}}\rangle\, e^{i\mathbf{k}\cdot\mathbf{x}} \quad . \tag{D5}$$

By setting the auxiliary parameter λ to unity, it is reduced to the density power spectrum. For Eq. (D4), the zeroth order moment is the redshift-space density power spectrum, $\Xi_s^{(0)}(\mathbf{k}) = P_{gg}^s(\mathbf{k})$. The first order derivative yields the density-velocity cross-correlation $\Xi_s^{(1)}(\mathbf{k}) = -2P_{gu}^s(\mathbf{k})$. The second derivative yields the unnormalized velocity dispersion $\Xi_s^{(2)}(\mathbf{k})$. We utilize this generic relation to compute $P_{10}^s(\mathbf{k})$, and evaluate the first derivative using the finite-difference strategy implemented in Ref. [47].

The velocity auto-correlation $P_{uu}^s(\mathbf{k})$ is not a pairwise velocity moment, and therefore cannot be obtained directly from Eq. (D4). Instead, we directly derive its analytical expressions within the same framework, and use the well-prepared mathematical result presented in literature. Theoretically, $\Xi_s^{(2)}$ is a better velocity statistic compared to $P_{uu}^s(\mathbf{k})$, since it is particularly convenient to compute under this framework and both are expected to encode same cosmological information. However, the measurement of velocity dispersion $\Xi_s^{(2)}$ requires the squared velocity field, which inevitably amplifies the observational noise and systematic effects. Therefore, we retain to use P_{uu}^s in our cosmological analysis.

2. Lagrangian bias expansion and model specifications

On large scales, the equivalence principle guarantees that the only measurable quantity in the non-relativistic limit is the second-order derivative of the gravitational potential $\partial_i\partial_j\Phi$, therefore the galaxy distribution is a general function $F[\partial_i\partial_j\Phi](\mathbf{q})$, where we have adopted the Lagrangian prescription. In Lagrangian space, the galaxy overdensity field is expanded as

$$\begin{aligned} 1+\delta_g^L(\mathbf{q}) &= F[\partial_i\partial_j\Phi](\mathbf{q}) \\ &= 1 + b_1\delta(\mathbf{q}) + b_2\left[\delta^2(\mathbf{q}) - \sigma^2\right] + b_s\left[s^2(\mathbf{q}) - \frac{2}{3}\sigma^2\right] + b_3\left[\delta^3(\mathbf{q}) - 3\sigma_L^2\delta(\mathbf{q})\right] + \varepsilon(\mathbf{q}) \end{aligned} \tag{D6}$$

where $\delta(\mathbf{q}) \equiv \nabla^2\Phi$ is the matter density contrast, $s^2 \equiv s_{ij}s_{ij}$ is the scalar tidal operator and the tidal tensor is $s_{ij} = \left(\partial_i\partial_j\,\nabla^{-2} - \frac{1}{3}\delta_{ij}^K\right)\delta$. The stochastic field ε accounts for fluctuations in the biased tracers that are uncorrelated with the long wavelength deterministic operators. Higher-order and higher-derivative bias operators are omitted here. We advect the Lagrangian basis fields $\mathcal{O}_i \in \{1, \delta, \delta^2-\sigma^2, s^2-\frac{2}{3}\sigma^2, \delta^3-3\sigma^2\delta\}$ into redshift-space according to

$$\mathcal{O}_i^s(\mathbf{s}, z) = \int_{\mathbf{q}} \mathcal{O}_i(\mathbf{q})\,\delta^D(\mathbf{s}-\mathbf{q}-\mathbf{\Psi}^s(\mathbf{q},z)) \quad , \tag{D7}$$

$$\mathcal{O}_i^s(\mathbf{k}, z) = \int_{\mathbf{q}} \mathcal{O}_i(\mathbf{q})\,e^{-i\mathbf{k}\cdot(\mathbf{q}+\mathbf{\Psi}^s(\mathbf{q},z))} \quad . \tag{D8}$$

The redshift-space galaxy power spectrum can then be expressed as

$$P_{gg}^s(k,\mu_{\mathbf{k}}) = \sum_{ij} b_i b_j P_{ij}^s(k,\mu_{\mathbf{k}}) + \sum_{mn}\alpha_{mn}k^{2n}\mu_{\mathbf{k}}^{2m}P_L(k) + \sum_{mn}\beta_{mn}k^{2n}\mu_{\mathbf{k}}^{2m} \tag{D9}$$

with $\mu_{\mathbf{k}} = \hat{\mathbf{k}}\cdot\hat{n}$. The α_{mn} terms are counterterms arising from the effective field theory that parameterize unresolved short-scale effects, while the β_{mn} terms describe stochastic contributions. The basis spectra are

$$P_{ij}^s(k,\mu_{\mathbf{k}}) = \int_{\mathbf{q}} \langle \mathcal{O}_i(\mathbf{q}_1)\mathcal{O}_j(\mathbf{q}_2)e^{i\mathbf{k}\cdot\Delta\mathbf{\Psi}^s}\rangle\, e^{i\mathbf{k}\cdot\mathbf{q}} \tag{D10}$$

and they are evaluated using the cumulant expansion. Using Lagrangian perturbation expansion, the displacement field and velocity field are $\mathbf{\Psi} = \sum_m \mathbf{\Psi}^{(m)}$ and $\mathbf{u} = \sum_m u^{(m)}\hat{n}$, with $u^{(m)} = fm\mathbf{\Psi}^{(m)}\cdot\hat{n}$. The leading long-wavelength contributions are retained in the exponent, while the remaining higher order contributions are expanded in the prefactor. Further implementation details are given in Refs. [59, 61]. Then, we obtain the density power spectrum, provided the galaxy bias parameters are given, and therefore obtain the density-velocity cross-power spectrum from the relation of Eq. (D4).

Similarly, the velocity auto-power spectrum is modeled as

$$P_{uu}^s(k,\mu_{\mathbf{k}}) = \int_{\mathbf{q}} \langle\, [1+b_1\delta(\mathbf{q}_1)]\,[1+b_1\delta(\mathbf{q}_2)]\,u(\mathbf{q}_1)u(\mathbf{q}_2)e^{i\mathbf{k}\cdot\Delta\mathbf{\Psi}^s}\rangle\, e^{i\mathbf{k}\cdot\mathbf{q}}\,. \tag{D11}$$

Here, we retain only the linear density bias b_1 for the tracer density weighting in velocity correlations. In the Lagrangian bias expansion up to one-loop order, the contributions from higher order density bias to velocity auto-correlation vanish. As we show below, the one-loop model together with the linear bias is sufficient for current precision in peculiar velocity measurements. By denoting $\Delta^s \equiv \mathbf{k}\cdot\Delta\mathbf{\Psi}^s$, the cumulant expansion gives

$$\begin{aligned}
&\langle\,[1+b_1\delta_1]\,[1+b_1\delta_2]\;u_1u_2\,e^{i\Delta^s}\,\rangle \\
&= e^{-\frac{1}{2}\langle\Delta^s\Delta^s\rangle+\cdots}\Big\{\langle u_1u_2\rangle_c + i\langle\Delta^s\,u_1u_2\rangle_c - \langle\Delta^s\,u_1\rangle_c\langle\Delta^s\,u_2\rangle_c \\
&\quad + b_1\Big[\langle(\delta_1+\delta_2)\,u_1u_2\rangle_c + i\langle u_1u_2\rangle_c\langle\Delta^s(\delta_1+\delta_2)\rangle_c + i\langle\Delta^s u_2\rangle_c\langle u_1\delta_2\rangle_c + i\langle\Delta^s u_1\rangle_c\langle u_2\delta_1\rangle_c\Big] \\
&\quad + b_1^2\Big[\langle u_2\delta_1\rangle_c\langle u_1\delta_2\rangle_c + \langle u_1u_2\rangle_c\langle\delta_1\delta_2\rangle_c\Big] \quad + \cdots\Big\}
\end{aligned} \tag{D12}$$

where $\langle\cdots\rangle_c$ denote the cumulants. On the right side of the equality, the first line is the matter velocity auto-correlation, which is the dominant contribution to the galaxy velocity auto-correlation at $s \gtrsim 10\,\mathrm{Mpc}/h$. The second and third lines arise from the density weighting of the velocity field, which only becomes significant at small scales. In fact, the scale cutoff already restricts the analysis to sufficiently linear regimes, and biased terms are negligible. Nevertheless, we use exact expressions up to one-loop order for consistent modeling as the density field.

To make use of the available theoretical results (e.g., [83, 84]) for the velocity auto power spectrum calculation, we need to express these correlators of velocity u_1 and u_2 in terms of correlators of the displacement $\mathbf{\Psi}$ or correlators of the pairwise velocity $u_{21} = u_2 - u_1$. For instance, we have $\langle\Delta^s\,u_1\rangle_c\langle\Delta^s\,u_2\rangle_c = -\frac{1}{4}\langle\Delta^s\,u_{21}\rangle_c^2 \sim \langle\Delta^s\,\Delta\rangle_c^2$ schematically, which straightforwardly corresponds to the available results in the literature. Some of the terms can not be expressed in terms of the pairwise velocity, while they are still directly obtained by tedious calculations. Up to one-loop order, the specific expressions are given below. **(1)** The matter velocity field contains the three terms, $\langle u_1u_2\rangle_c$, $\langle\Delta^s\,u_1u_2\rangle_c$ and $\langle\Delta^s\,u_1\rangle_c\langle\Delta^s\,u_2\rangle_c$.

$$\langle u_1u_2\rangle_c = f^2\langle\Psi^{(1)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c + 4f^2\langle\Psi^{(2)}_{1\hat{n}}\Psi^{(2)}_{2\hat{n}}\rangle_c + 6f^2\langle\Psi^{(1)}_{1\hat{n}}\Psi^{(3)}_{2\hat{n}}\rangle_c \tag{D13}$$

$$\langle\Psi^{(1)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c = \frac{1}{3}\left[\xi^0_{-2}(q) + \xi^2_{-2}(q)\right] - \xi^2_{-2}(q)\,(\hat{n}\cdot\hat{q})^2 \tag{D14}$$

$$\langle\Psi^{(2)}_{1\hat{n}}\Psi^{(2)}_{2\hat{n}}\rangle_c = \frac{3}{98}\left\{\Xi^0_{-2}\left[Q_1\right] + \Xi^2_{-2}\left[Q_1\right]\right\} - \frac{9}{98}\Xi^2_{-2}\left[Q_1\right]\,(\hat{n}\cdot\hat{q})^2 \tag{D15}$$

$$\langle\Psi^{(1)}_{1\hat{n}}\Psi^{(3)}_{2\hat{n}}\rangle_c = \frac{5}{63}\left\{\Xi^0_{-2}\left[R_1\right] + \Xi^2_{-2}\left[R_1\right]\right\} - \frac{5}{21}\Xi^2_{-2}\left[R_1\right]\,(\hat{n}\cdot\hat{q})^2 \tag{D16}$$

$$\langle\Delta^s\,u_1u_2\rangle_c = f^2\langle\Delta^{s(2)}\Psi^{(1)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c + 4f^2\,\langle\Delta^{s(1)}\Psi^{(2)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c \tag{D17}$$

$$\begin{aligned}
\langle\Delta^{s(2)}\Psi^{(1)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c &= \left[K\cdot\hat{q} + fk\mu_{\mathbf{k}}(\hat{n}\cdot\hat{q})\right]\,V_1' + \left[(K\cdot\hat{n})(\hat{n}\cdot\hat{q}) + fk\mu_{\mathbf{k}}(\hat{n}\cdot\hat{q})\right]\,(V_2' + V_3') \\
&\quad + (\hat{n}\cdot\hat{q})^2\left[K\cdot\hat{q} + fk\mu_{\mathbf{k}}(\hat{n}\cdot\hat{q})\right]\,T'
\end{aligned} \tag{D18}$$

$$\langle\Delta^{s(1)}\Psi^{(2)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c = (K\cdot\hat{q})\,V_1'' + (\hat{n}\cdot\hat{q})(K\cdot\hat{n})\,(V_2'' + V_3'') + (\hat{n}\cdot\hat{q})^2(K\cdot\hat{q})\,T'' \tag{D19}$$

$$V_1'(q) = -\frac{3}{35}\Xi^1_{-3}\left[R_1 + 2R_2\right] - \frac{1}{5}T'(q) \tag{D20}$$

$$V_3'(q) = -\frac{3}{70}\Xi^1_{-3}\left[-3R_1 + 4R_2\right] - \frac{1}{5}T'(q) \tag{D21}$$

$$V_2'(q) = V_3'(q) \tag{D22}$$

$$T'(q) = \frac{3}{7}\,\Xi^3_{-3}\left[R_1 + 2R_2\right] \tag{D23}$$

$$V_1''(q) = -\frac{3}{70}\Xi^1_{-3}\left[R_1 + 2R_2 + Q_1 + 2Q_2\right] - \frac{1}{5}T''(q) \tag{D24}$$

$$V_2''(q) = -\frac{3}{70}\Xi^1_{-3}\left[-4R_1 + 2R_2 + Q_1 + 2Q_2\right] - \frac{1}{5}T''(q) \tag{D25}$$

$$V_3''(q) = -\frac{3}{70}\Xi^1_{-3}\left[R_1 + 2R_2 - 4Q_1 + 2Q_2\right] - \frac{1}{5}T''(q) \tag{D26}$$

$$T''(q) = \frac{3}{14}\Xi^3_{-3}\left[Q_1 + 2Q_2 + R_1 + 2R_2\right] \tag{D27}$$

$$\langle \Delta^s\, u_1\rangle_c\langle \Delta^s\, u_2\rangle_c = -\frac{1}{4}\langle \Delta^{s(1)}\, u_{21}^{(1)}\rangle_c\langle \Delta^{s(1)}\, u_{21}^{(1)}\rangle_c \tag{D28}$$

$$f^{-1}\langle \Delta^{s(1)}\, u_{21}^{(1)}\rangle_c = \frac{2}{3}\left[\xi^0_{-2}(0) - \xi^0_{-2}(q) - \xi^2_{-2}(q)\right](K\cdot\hat{n}) + 2\xi^2_{-2}(q)(K\cdot\hat{q})(\hat{n}\cdot\hat{q}) \tag{D29}$$

(2) The matter-bias cross-correlation terms are $\langle(\delta_1+\delta_2)\, u_1u_2\rangle_c$, $\langle u_1u_2\rangle_c\langle\Delta^s(\delta_1+\delta_2)\rangle_c$ and $\langle\Delta^s u_2\rangle_c\langle u_1\delta_2\rangle_c$.

$$\langle(\delta_1+\delta_2)\, u_1u_2\rangle_c = 4f^2\left[\langle\delta_1\,\Psi^{(1)}_{1\hat{n}}\Psi^{(2)}_{2\hat{n}}\rangle_c + \langle\delta_1\,\Psi^{(2)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c\right] \tag{D30}$$

$$\langle\delta_1\,\Psi^{(1)}_{1\hat{n}}\Psi^{(2)}_{2\hat{n}}\rangle_c + \langle\delta_1\,\Psi^{(2)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c = \frac{1}{7}\Xi^0_{-2}[R_1+R_2+Q_5] + \frac{1}{7}\Xi^2_{-2}[R_1+R_2+Q_5] - \frac{3}{7}\Xi^2_{-2}[R_1+R_2+Q_5](\hat{n}\cdot\hat{q})^2 \tag{D31}$$

$$\langle u_1u_2\rangle_c\langle\Delta^s(\delta_1+\delta_2)\rangle_c = 2f^2\,\langle\Psi^{(1)}_{1\hat{n}}\Psi^{(1)}_{2\hat{n}}\rangle_c\langle\Delta^{s(1)}\delta_1\rangle_c \tag{D32}$$

$$\langle\Delta^s u_2\rangle_c\langle u_1\delta_2\rangle_c = -\frac{1}{2}\langle\Delta^{s(1)}u_{21}^{(1)}\rangle_c\langle u_{21}^{(1)}\delta_2\rangle_c \tag{D33}$$

(3) The bias auto-correlation terms are $\langle u_2\delta_1\rangle_c\langle u_1\delta_2\rangle_c$ and $\langle u_1u_2\rangle_c\langle\delta_1\delta_2\rangle_c$.

$$\langle u_2\delta_1\rangle_c\langle u_1\delta_2\rangle_c = -\langle u_{21}^{(1)}\delta_1\rangle_c^2 \tag{D34}$$

$$\langle u_1u_2\rangle_c\langle\delta_1\delta_2\rangle_c = \langle u_1^{(1)}u_2^{(1)}\rangle_c\langle\delta_1\delta_2\rangle_c \tag{D35}$$

Here, we have denote $\Psi^{(n)}_{1\hat{n}} = \mathbf{\Psi}^{(n)}(\mathbf{q}_1)\cdot\hat{n}$, $\Psi^{(n)}_{2\hat{n}} = \mathbf{\Psi}^{(n)}(\mathbf{q}_2)\cdot\hat{n}$, and $K = \mathbf{k} + fk\mu_{\mathbf{k}}\hat{n}$ [47, 61]. We have used the shorthand for integration

$$\Xi^\ell_n[\mathcal{F}](q) \equiv \frac{1}{2\pi^2}\int_0^\infty k^2\mathrm{d}k\, j_\ell(kq)\, k^n\mathcal{F}(k)$$

and the generalized correlation function is defined as $\xi^\ell_n(q) \equiv \Xi^\ell_n[P_L](q)$ accordingly [85]. Here, $P_L(k)$ is the linear matter power spectrum, the input of the perturbation calculation. The $Q(k)$ and $R(k)$ functions are presented in Refs. [83, 84].

Combining the results above, we obtain theoretical models for the redshift-space power spectra P^s_{gg}, P^s_{gu}, and P^s_{uu}. These spectra depend on the cosmological parameters encoded in the linear matter power spectrum $P_L(k)$, the galaxy bias parameters $\{b_1, b_2, b_s, b_3, \cdots\}$, and the associated counterterms and stochastic contributions. However, current observations are limited by survey volume, and peculiar velocities are particularly affected by substantial measurement noise. Consequently, the measurements can not constrain the full set of parameters. We therefore simplify the model by discarding the terms with negligible contribution, while retaining the flexibility of the model.

- Since the measurements are only sensitive to the overall amplitude for both redshift space distortions and peculiar velocities, we only vary $f\sigma_8$ parameter of the cosmological input. Other cosmological parameters, insensitive to current analysis, are fixed to the fiducial values of the Planck18 cosmology [45].
- The galaxy density auto-power spectrum retains three parameters, $\{b_1, b_2, b_s\}$. Here, we neglect the higher order bias and discard all the counterterms and stochastic terms, which are subdominant compared to the statistical uncertainties over the conservative scale cutoff.
- The galaxy velocity auto-power spectrum retains only the linear density bias, $\{b_1\}$. In current measurement precision, the constraint power is dominated by the matter velocity field, whereas corrections arising from density weighting and stochastic effects become important only on smaller scales.
- The density-velocity cross-power spectrum requires an additional parameter, $\{\alpha_v\}$, a counterterm that parameterizes unresolved short-scale contributions to the pairwise velocity as $P^s_{gu}(k,\mu_{\mathbf{k}}) \supset i\alpha_v k\mu_{\mathbf{k}}P_L(k)$.

The resulting fitting parameter sets are therefore $\{f\sigma_8, b_1\}$ for velocity-only analysis, $\{f\sigma_8, b_1, b_2, b_s\}$ for density-only analysis, and $\{f\sigma_8, b_1, b_2, b_s, \alpha_v\}$ for joint analysis of galaxy velocity and density.

We compute the linear matter power spectrum $P_L(k)$ using the Boltzmann solver `CLASS` [86]. Then we evaluate the redshift-space galaxy density power spectrum P^s_{gg} using the `lpt_rsd_fftw` module [87] of the public `velocileptors` package [60, 61]. The galaxy density-velocity cross power spectrum P^s_{gu} is subsequently obtained by taking the finite difference derivative of the density power spectrum with respect to the linear growth rate f, following the implementation in `velocisaurus`. To compute the velocity auto power spectrum P^s_{uu}, we extend the `velocisaurus` implementation [88] by incorporating the analytical expressions derived above. In our publicly released code [89], we provide the full implementation of the redshift-space statistics calculation, together with the all the modifications mentioned above, using in this work.

3. Correlation function multipoles and density normalization

The Lagrangian bias expansion described above provides predictions for the redshift-space power spectra of galaxy density field $\delta_g(\mathbf{s})$ and density-weighted velocity field (or the momentum field) $p_g(\mathbf{s}) = [1+\delta_v(\mathbf{s})]\,u(\mathbf{s})$, where δ_v is the overdensity of the velocity sample. We obtain the correlation function multipoles in configuration space by applying Hankel transforms to the Fourier space results. For $X, Y \in \{\delta_g, p_g\}$, the anisotropic correlation functions are $\xi^{XY}(s,\mu) = \langle X(\mathbf{s}_1)Y(\mathbf{s}_2)\rangle\big|_{\mathbf{s}=\mathbf{s}_1-\mathbf{s}_2}$, and become

$$\xi_\ell(s) = \frac{2\ell+1}{2}\int_{-1}^{1} d\mu\,\mathcal{L}_\ell(\mu)\xi(s,\mu)\;, \quad \xi(s,\mu) = \sum_\ell \mathcal{L}_\ell(\mu)\xi_\ell(s)$$

by expanding in terms of Legendre polynomials $\mathcal{L}_\ell(\mu)$. However, the velocity estimators are normalized by the pair counts of the data samples. Consequently, their theoretical predictions require an additional normalization by the density cross-correlation of the two samples, as shown in Eqs. (A7, A8 & A9). In the following, we describe our treatment for this density normalization in the correlation function modeling.

The density normalization can be applied directly as a pointwise division when the correlation functions are evaluated as functions of (s,μ). However, to maintain the computational efficiency, the building block of predictions is multipole, function of (s,ℓ), coupling the contributions from different angular pairs. For consistent implementation, we need to express the density normalization in terms of a series of multipoles. The normalized velocity statistics can be written in a unified form as

$$\begin{aligned}
\xi^{LL'}(\mathbf{s}) &= \frac{\langle[1+\delta(\mathbf{s}_1)][1+\delta(\mathbf{s}_2)]u^L(\mathbf{s}_1)u^{L'}(\mathbf{s}_2)\rangle}{\langle[1+\delta(\mathbf{s}_1)][1+\delta(\mathbf{s}_2)]\rangle} \\
&= \frac{\xi^{p_1p_2}(\mathbf{s})}{1+\xi^{\delta_1\delta_2}(\mathbf{s})} \\
&= \xi^{p_1p_2}(\mathbf{s}) - \xi^{p_1p_2}(\mathbf{s})\xi^{\delta_1\delta_2}(\mathbf{s}) + \xi^{p_1p_2}(\mathbf{s})\xi^{\delta_1\delta_2}(\mathbf{s})\xi^{\delta_1\delta_2}(\mathbf{s}) + \cdots
\end{aligned}$$

Here, $\xi^{\delta_1\delta_2}(\mathbf{s})$ is the ordinary galaxy density cross-correlation, and $\xi^{p_1p_2}(\mathbf{s})$ is the unnormalized density-weighted velocity correlation. The choice $(L,L') = (1,1)$ yields the velocity auto-correlation $\xi^{uu}(\mathbf{s})$, while $(L,L') = (0,1)$ yields the density-velocity cross-correlation $\xi^{gu}(\mathbf{s})$. In the third equality, we utilize the approximation $\xi^{\delta_1\delta_2}(\mathbf{s}) \ll 1$ on large scales, e.g., at $z \simeq 0.1$, there is $\xi^{\delta_m\delta_m}(r) \lesssim 0.1$ on scales $r \gtrsim 20\,\mathrm{Mpc}/h$, therefore we are able to expand the denominator in a Taylor series. In multipole space, the expansion of the normalized velocity statistics becomes

$$\xi_\ell^{LL'}(s) = \xi_\ell^{p_1p_2}(s) + \eta_\ell^{(1)}(s) + \eta_\ell^{(2)}(s) + \cdots$$

where $\xi_\ell^{p_1p_2}(s)$ is the predicted multipoles of the unnormalized velocity statistic introduced above, and $\eta_\ell^{(n)}(s)$ denotes the n-th-order density-normalization correction to the ℓ-th multipole. The first two order corrections are

$$\begin{aligned}
\eta_\ell^{(1)}(s) &= -\frac{2\ell+1}{2}\int_{-1}^{1} d\mu\;\mathcal{L}_\ell(\mu)\;\xi^{p_1p_2}(s,\mu)\xi^{\delta_1\delta_2}(s,\mu) \\
&= -\frac{2\ell+1}{2}\sum_{\ell_1,\ell_2}\int_{-1}^{1} d\mu\;\mathcal{L}_\ell(\mu)\mathcal{L}_{\ell_1}(\mu)\mathcal{L}_{\ell_2}(\mu)\;\xi_{\ell_1}^{p_1p_2}(s)\xi_{\ell_2}^{\delta_1\delta_2}(s) \\
&= -(2\ell+1)\sum_{\ell_1,\ell_2}\begin{pmatrix} l & l_1 & l_2 \\ 0 & 0 & 0\end{pmatrix}^2\;\xi_{\ell_1}^{p_1p_2}(s)\xi_{\ell_2}^{\delta_1\delta_2}(s) \\
\eta_\ell^{(2)}(s) &= \frac{2\ell+1}{2}\int_{-1}^{1} d\mu\;\mathcal{L}_\ell(\mu)\;\xi^{p_1p_2}(s,\mu)\xi^{\delta_1\delta_2}(s,\mu)\xi^{\delta_1\delta_2}(s,\mu) \\
&= \frac{2\ell+1}{2}\sum_{\ell_1\ell_2\ell_3}\int_{-1}^{1} d\mu\;\mathcal{L}_\ell(\mu)\mathcal{L}_{\ell_1}(\mu)\mathcal{L}_{\ell_2}(\mu)\mathcal{L}_{\ell_3}(\mu)\;\xi_{\ell_1}^{p_1p_2}(s)\xi_{\ell_2}^{\delta_1\delta_2}(s)\xi_{\ell_3}^{\delta_1\delta_2}(s) \\
&= (2\ell+1)\sum_{\ell_1\ell_2\ell_3}\left[\sum_L(2L+1)\;\begin{pmatrix} l & l_1 & L \\ 0 & 0 & 0\end{pmatrix}^2\begin{pmatrix} l_2 & l_3 & L \\ 0 & 0 & 0\end{pmatrix}^2\right]\xi_{\ell_1}^{p_1p_2}(s)\xi_{\ell_2}^{\delta_1\delta_2}(s)\xi_{\ell_3}^{\delta_1\delta_2}(s)
\end{aligned}$$

Here, multiplication of correlation functions in angular space produces a mode coupling in multipole space. We have transformed the Legendre polynomials integrals into the Gaunt integrals of Wigner $3j$ symbols by

$$\frac{1}{2}\int_{-1}^{1}\mathcal{L}_{\ell_1}(\mu)\mathcal{L}_{\ell_2}(\mu)\mathcal{L}_{\ell_3}(\mu)d\mu = \begin{pmatrix} l_1 & l_2 & l_3 \\ 0 & 0 & 0\end{pmatrix}^2 \quad ,$$

$$\frac{1}{2}\int_{-1}^{1}\mathcal{L}_{\ell_1}(\mu)\mathcal{L}_{\ell_2}(\mu)\mathcal{L}_{\ell_3}(\mu)\mathcal{L}_{\ell_4}(\mu)d\mu=\sum_{L}(2L+1)\begin{pmatrix}l_1 & l_2 & L\\ 0 & 0 & 0\end{pmatrix}^2\begin{pmatrix}l_3 & l_4 & L\\ 0 & 0 & 0\end{pmatrix}^2 \quad .$$

In the main analysis and the associated validations, we adopt corrections of the density normalization up to the second order, and truncate the summation of Wigner $3j$ symbols up to 4 for $L, \ell_1, \ell_2, \cdots$. As demonstrated in the following section, this treatment is sufficient to recover unbiased cosmological constraints from the velocity statistics.

4. Validation based on FLAMINGO simulations

In this section, we test the Lagrangian bias expansion theory together with all treatments described above, using three cosmological variants of the FLAMINGO simulations suite [70–72] to validate the model performance.

FLAMINGO is a suite of large-volume hydrodynamical simulations designed for large-scale structure and galaxy cluster surveys [70–72], with variants differing in subgrid implementation, cosmology, resolution, and box size. We use three intermediate-resolution simulations, each with a comoving volume of $1\,\mathrm{Gpc}^3$, and containing 1800^3 dark matter particles, 1800^3 baryon particles, and 1000^3 neutrino particles. Subhalos and their properties are identified using the HBT-HERONS structure finder, provided with the SOAP catalogs [73–75]. We use the snapshot at redshift $z = 0.1$, and treat all luminous subhalos in these catalogs as galaxies. The baseline simulation and two cosmological variants considered in this work are

- `L1_m9`: The cosmological parameter of the baseline cosmology are taken from the maximum posterior likelihood values from the Dark Energy Survey Year three [90], with $f\sigma_8 = 0.446$ at $z = 0.1$.
- `Planck`: The simulation adopts the best-fit ΛCDM cosmology from Planck 2018 results [45], with $f\sigma_8 = 0.448$ at $z = 0.1$.
- `LS8`: The simulation adopts a lower matter power spectrum amplitude with $S_8 = 0.766$, which is adjusted to better match galaxy clustering and cosmic shear observations, with $f\sigma_8 = 0.420$ at $z = 0.1$.

For each simulation, we construct five subsamples of galaxies selected by their stellar mass, spanning a range of clustering properties. Each sample has a fixed number density, $\bar{n}_g = 4\times10^{-3}\,h^3\mathrm{Mpc}^{-3}$, approximately matching that of the MGS galaxies. We apply redshift space distortions along a fixed line-of-sight direction and add white noise with variance $\sigma_v^2 = 4.2\times10^7\,(\mathrm{km/s})^2$ to the peculiar velocities, matching the mean velocity dispersion of the `massE` catalog. In the simulations, the same galaxy sample serves as both the density and velocity tracers, whereas the number density of the observed `massE` velocity catalog is approximately half that of the MGS density catalog. The effective velocity shot noise in the simulations is therefore approximately half that in the observational data. It results in smaller statistical uncertainties, and provides a more stringent test of the theoretical velocity statistics. Together, the three cosmologies and five stellar-mass-selected samples provide 15 test cases spanning a range of cosmological and tracer properties.

To reduce the computational cost, the covariance matrix is estimated as $\mathbf{Cov} = \mathbf{Cov}_{\mathrm{sub}}/N_{\mathrm{sub}}$, instead of using jackknife resampling. Here, $\mathbf{Cov}_{\mathrm{sub}}$ is the covariance matrix from simulation sub-box with equal volume, and $N_{\mathrm{sub}} = 4^3$ is the number of sub-boxes we divide. To reduce the sample variance, we take the x, y, and z axes in turn as the line-of-sight direction, and average three estimates. An example covariance matrix for the `L1_m9` simulation is shown in the left panel of of Fig. 27. Compared to the data covariance using the jackknife resampling (Fig. 30), the simulation covariance exhibits stronger correlations among different multipoles. This difference is not primarily caused by the estimation method, as we have verified that jackknife resampling to the simulations produces a similar correlation structure. Instead, we do not add any observational noise except the white noise to the simulated galaxies, therefore the simulation covariance is much more dominated by the cosmic variance, which coherently correlates the different multipoles. Fig. 26 presents an example fit to the data vector $\{\xi^{gg}_{0,2}, \xi^{gu}_{1}, \xi^{uu}_{0,2}\}$. The simulation measurements have substantially smaller error bars in velocity statistics than the observational measurements, making this a more stringent validation of the theoretical model.

The right panel of Fig. 27 summarizes the marginalized constraints and MAP estimates of $f\sigma_8$ obtained from the joint analysis, velocity-only analysis, and density-only analysis for all 15 subsamples. For both the velocity-only data vector $\{\xi^{uu}_{0,2}\}$ and the joint data vector $\{\xi^{gg}_{0,2}, \xi^{gu}_{1}, \xi^{uu}_{0,2}\}$, the inferred values are consistent with the true values within the statistical uncertainties, and the joint analysis generally yields tighter constraints. The density-only fits also recover MAP values close to the true cosmologies, although their marginalized means are systematically lower with differences of approximately 1σ. The discrepancy arises from the intrinsic degeneracy between $f\sigma_8$ and the galaxy bias parameters, since the density-only multipoles are sensitive to the structure growth rate by the redshift-space distortions, a higher-order effect in the galaxy density field. Figure 28 illustrates the origin of this difference

using the highest-stellar-mass subsample of the `L1_m9` simulation. The density-only posterior displays broad, curved degeneracies between $f\sigma_8$ and the galaxy bias parameters, particularly $(1 + b_1^L)\sigma_8$. Marginalizing over the extended nuisance parameter consequently shifts the one-dimensional posterior mean away from the MAP value. It therefore a projection effect associated with the non-Gaussian posterior geometry, rather than from a failure of the model to recover the true cosmology. Figure 29 presents the posterior distributions of $f\sigma_8$ and the leading order bias parameter b_1 for all 15 subsamples, demonstrating the intrinsic parameter degeneracy among different data combinations.

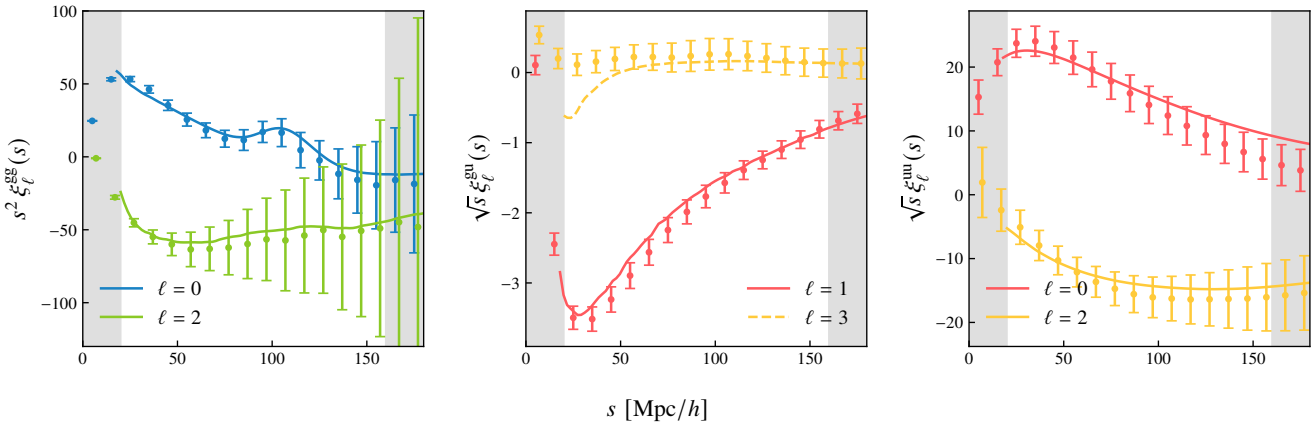


FIG. 26. **Validation of the Lagrangian bias expansion model using the FLAMINGO simulations.** Similar to Fig. 9, we jointly fit the Lagrangian bias model to data vector $\{\xi_0^{gg}, \xi_2^{gg}, \xi_1^{gu}, \xi_0^{uu}, \xi_2^{uu}\}$ measured from the highest-stellar-mass subsample of the `L1_m9` simulation, with $\chi^2/\mathrm{dof} = 0.48$. The results for the remaining 14 subsamples are similar and are omitted for brevity.

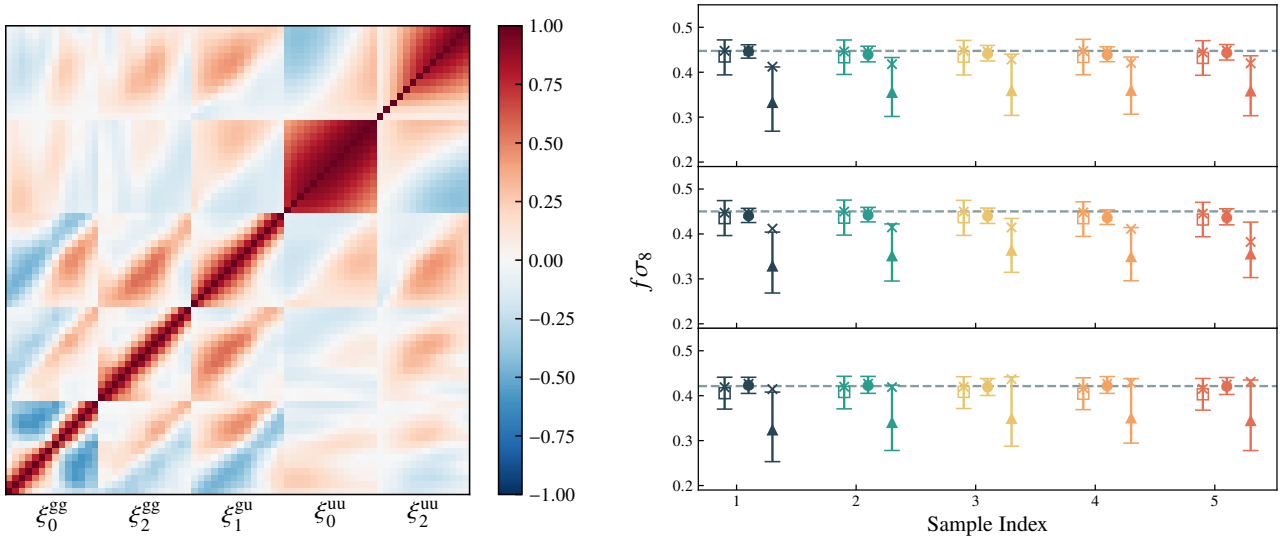


FIG. 27. **Validation of the Lagrangian bias expansion model using the FLAMINGO simulations. Left panel:** Covariance matrix of the data vector $\{\xi_0^{gg}, \xi_2^{gg}, \xi_1^{gu}, \xi_0^{uu}, \xi_2^{uu}\}$ measured from the highest-stellar-mass subsample of the `L1_m9` simulation. The covariance matrix is estimated by dividing the simulation volume into 4^3 sub-boxes and averaging the estimates obtained by taking the x, y, and z axes in turn as the line-of-sight direction. **Right panel**: Same as Fig. 7, we additionally show the $f\sigma_8$ constraints from the density-only analysis, denoted by triangular markers.

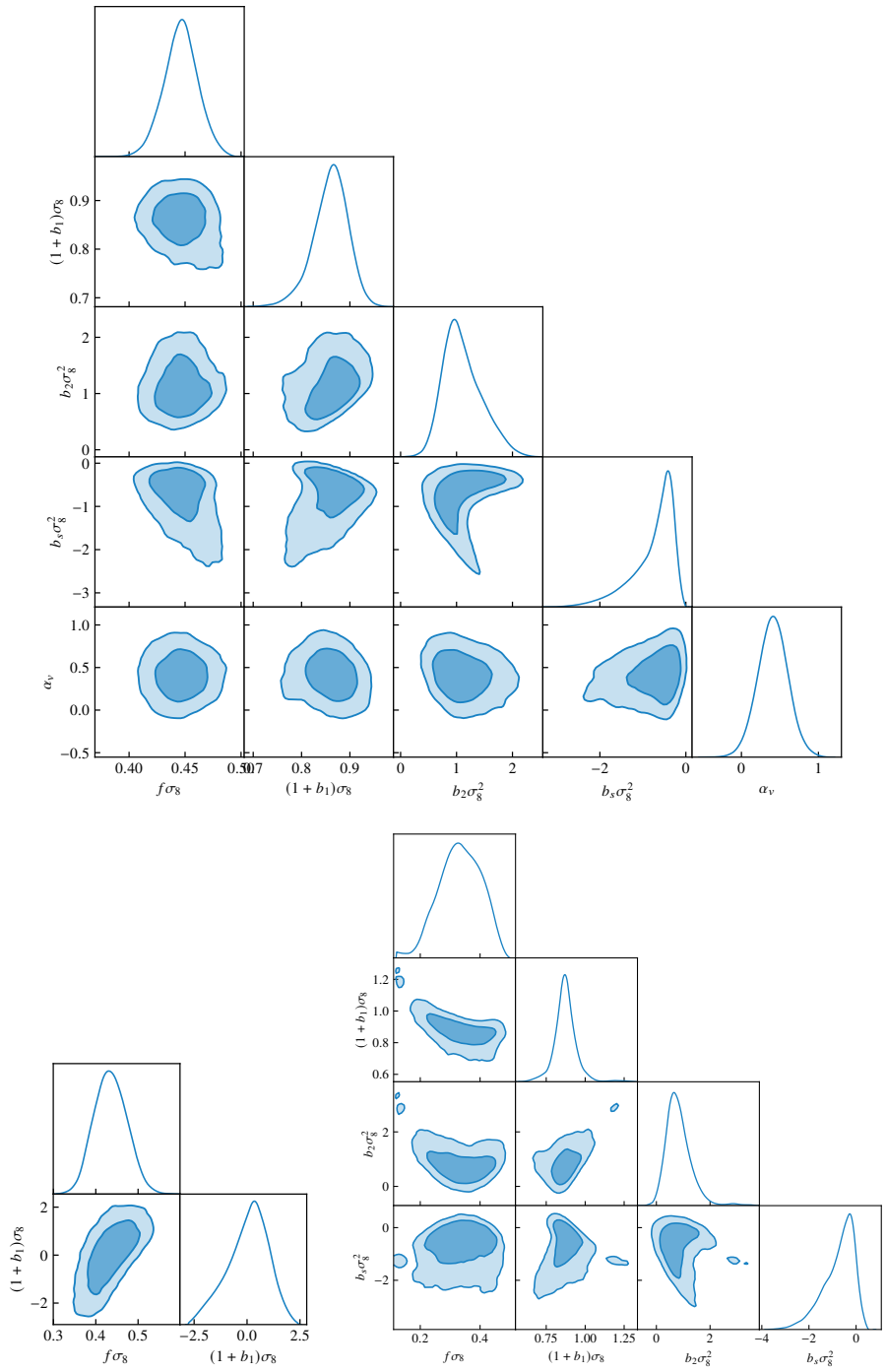


FIG. 28. **Validation of the Lagrangian bias expansion model using the FLAMINGO simulations.** We show the full posterior distributions of the model parameters inferred using the highest-stellar-mass subsample in the `L1_m9` simulation. **Top panel:** Joint analysis with the data vector $\{\xi_0^{gg}, \xi_2^{gg}, \xi_1^{gu}, \xi_0^{uu}, \xi_2^{uu}\}$. **Bottom left panel:** Velocity-only analysis with $\{\xi_0^{uu}, \xi_2^{uu}\}$. **Bottom right panel:** Density-only analysis with $\{\xi_0^{gg}, \xi_2^{gg}\}$.

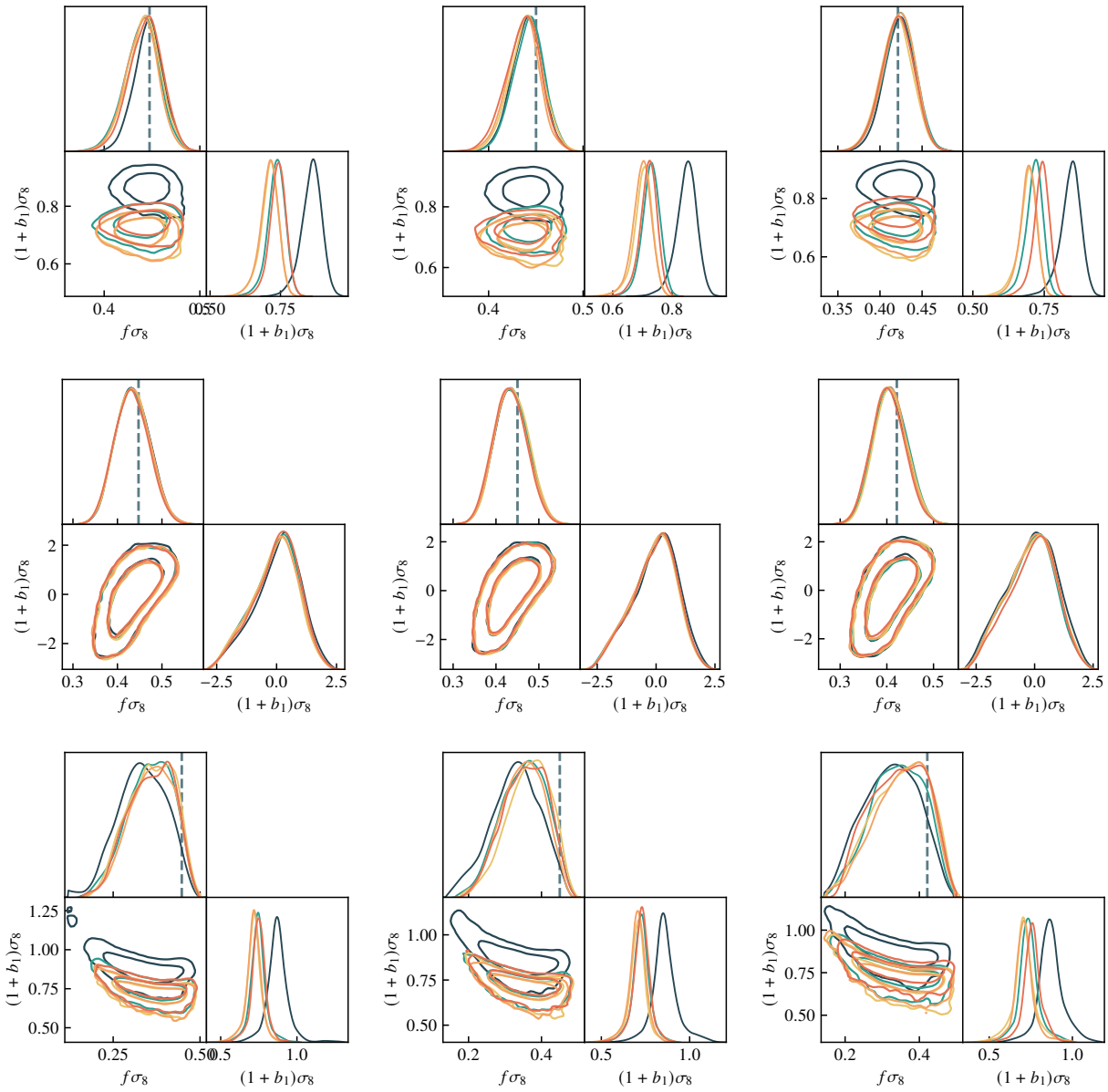


FIG. 29. **Validation of the Lagrangian bias expansion model using the FLAMINGO simulations.** We show the posterior distributions of $f\sigma_8$ and the linear bias parameter b_1, with dashed vertical lines indicating the true values of $f\sigma_8$. From left to right, the columns present results for the `L1_m9`, `Planck`, and `LS8` simulations, respectively. The five stellar-mass-selected subsamples are distinguished by color. **Top row**: Constraints from joint analysis of the velocity and density statistics. **Middle row**: Constraints from the velocity-only analysis. **Bottom row**: Constraints from the density-only analysis.

Appendix E: Auxiliary cosmological analysis for massE peculiar velocity

Fig. 30 shows the covariance matrix estimated using the fiducial choice of $N_{\rm JK} = 400$ jackknife regions, as described in Appendix. A. Substantial correlations occur between separation bins within each multipole, particularly for the velocity auto-correlation multipoles. Consequently, visually coherent differences between the measurements and the best-fit model do not necessarily indicate a poor fit when the full covariance is included. Fig. 31 compares the covariance matrices estimated with different numbers of jackknife resampling. If $N_{\rm JK}$ is too large, the small regions become correlated and the large-scale variance may be underestimated. As shown in the left panel of Fig. 31, the large-scale variances decrease as $N_{\rm JK}$ increase. Conversely, if $N_{\rm JK}$ is too small, the covariance matrix becomes noisy and the off-diagonal components are poorly resolved. As shown in the right panel of Fig. 31, small eigenvalues are increasingly suppressed as $N_{\rm JK}$ decreases, with $N_{\rm JK} = 81$ showing the largest deviation. We therefore compare $N_{\rm JK} = 169$, 400 in the parameter inference, considering the trade-off between these effects. As shown in Fig. 32, the posterior distributions of $f\sigma_8$ and the galaxy bias parameters are broadly consistent, and the marginalized constraints remain stable relative to their statistical uncertainties. Although modest offsets remain, they are small relative to the statistical uncertainties and do not affect our main conclusions. We therefore adopt $N_{\rm JK} = 400$ as the fiducial choice.

The potential bias in the jackknife variance increases with separation s, as in Fig. 31. To assess its impact, we repeat the analysis over the restricted fitting range $s \in [20, 120]\,{\rm Mpc}/h$, with the resulting constraints shown in Fig. 33. The constraints remain consistent with the fiducial results, and the bimodal structure persists. The inference is therefore robust to the fitting range.

In both the theoretical modeling and the simulation validation for the cosmological analysis, we have adopted the flat sky approximation, which assumes a fixed line-of-sight direction for all galaxy pairs. This approximation is accurate for pairs with small opening angles, but may become inadequate for velocity statistics at large separations [49]. In Fig. 35, we explicitly quantify the order-of-magnitude differences between multipoles with and without this approximation. We use flat sky measurement as the reference (upper right panel of Fig. 18), and compare them to that from the curved-sky measurements with massE-like redshift distribution (bottom right panel of Fig. 18). For the density-velocity cross-correlation ξ_ℓ^{gu}, the curved-sky geometry generates additional even multipoles, and modifies $\ell = 1$ and $\ell = 3$ components at $s \gtrsim 100{\rm Mpc}/h$. For the velocity auto-correlation ξ_ℓ^{uu}, the odd multipoles vanish because of the symmetry of the midpoint estimator, whereas the even multipoles are increasingly modified. These corrections become noticeable at $s \gtrsim 20\,{\rm Mpc}/h$ for $\ell = 0$, 2. Nevertheless, for both statistics, the curved-sky measurements closely follow their flat sky counterparts on small and intermediate scales, and appreciable deviations emerge only toward the large separations. At the upper limit of our fitting range, $s \sim 160\,{\rm Mpc}/h$, the fractional deviation of ξ_1^{gu} reaches maximum with $\Delta\xi_1^{gu} \,/\, \xi_1^{gu} \sim 15\%$, subdominant to the observational uncertainty $\sigma_{\xi_1^{gu}} \,/\, \xi_1^{gu} \sim 100\%$. Similarly, the largest deviation of ξ_0^{uu} is $\sim 15\%$, overwhelmed by the statistical fluctuation of $\sigma_{\xi_0^{uu}} \,/\, \xi_0^{uu} \sim 50\%$. In current measurements, we therefore conclude that the wide angle corrections are subdominant to the statistical noise, and they also cannot explain the suppressed amplitude of the velocity statistics discussed in the main text.

To quantify the contribution the mean velocity within each redshift slice, we apply the shuffle-subtraction procedure to the dark matter mocks described in Appendix C. Fig. 36 shows the resulting correlation function multipoles, together with the one-loop Lagrangian perturbation theory predictions. For this survey volume setup, these mean velocities over redshift slice dominate the large-scale power of the velocity statistics, and the shuffle-subtraction procedure effectively removes this contribution, particularly from the velocity auto-correlation ξ_ℓ^{uu}. Cross-correlations between the redshift mean and residual velocities can still contribute to the velocity auto-correlation after shuffle-subtraction. Nevertheless, the large-scale power is reduced to nearly zero, indicating that this contribution is subdominant for our concern suppressions. Fig. 34 shows the model parameter posteriors corresponding to the fit in Fig. 12. The inferred $f\sigma_8$ remains consistent with the fiducial result, while the bimodal posterior structure disappears.

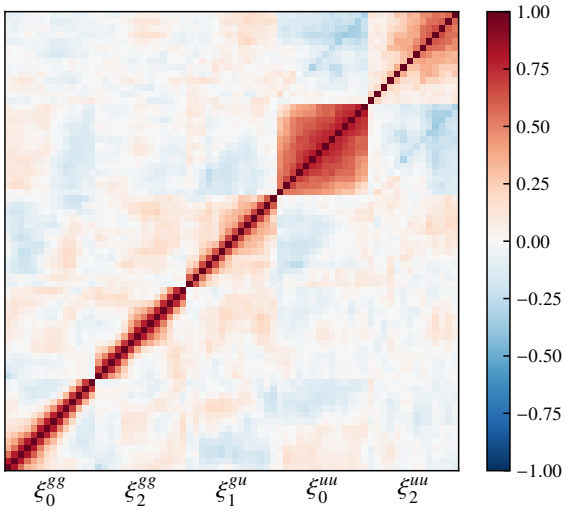


FIG. 30. **Covariance matrix of the galaxy statistics**, $\{\xi_0^{gg}, \xi_2^{gg}, \xi_1^{gu}, \xi_0^{uu}, \xi_2^{uu}\}$, measured from the `massE` velocity and MGS density catalogs. The correlation function multipoles are measured over $s \in [20, 160]\,\mathrm{Mpc}/h$ with a bin width of $\Delta s = 10\,\mathrm{Mpc}/h$. The covariance matrix is estimated from 400 jackknife subsamples, and the shown result is normalized by diagonal components for visualization.

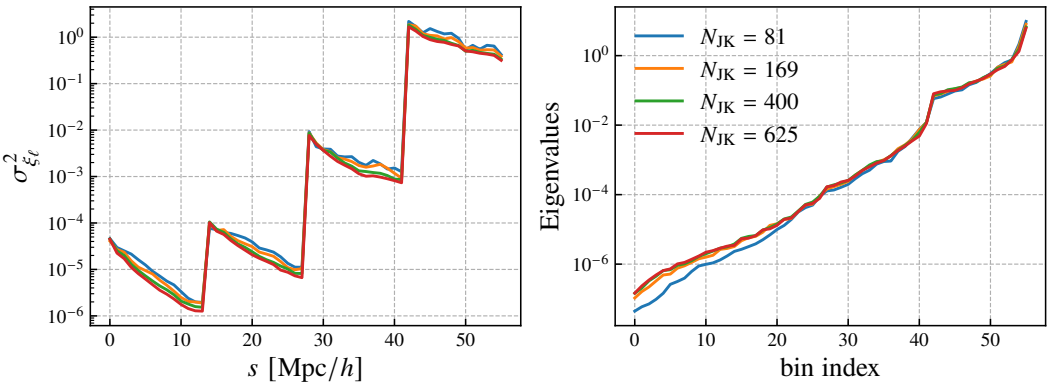


FIG. 31. **Left panel**: Diagonal components of the covariance matrix for data vector $\{\xi_0^{gg}, \xi_2^{gg}, \xi_1^{gu}, \xi_0^{uu}\}$ over $s \in [20, 160]\,\mathrm{Mpc}/h$, measured from `massE` velocity and MGS density catalogs. **Right panel**: Eigenvalues of corresponding covariance matrix. Different colors denote different numbers of jackknife regions, $N_{\rm JK}$.

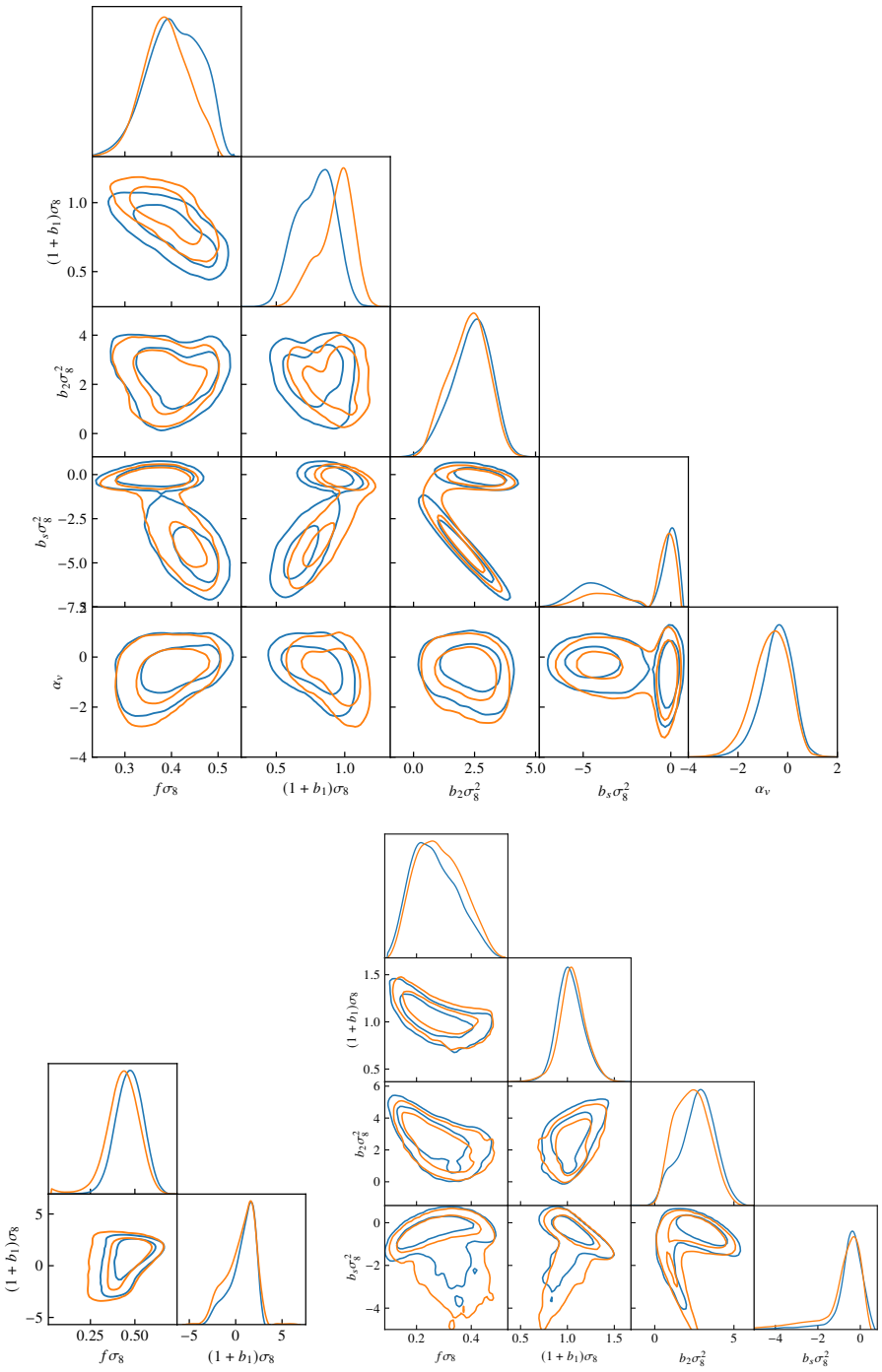


FIG. 32. **Posterior distributions of the model parameters** inferred from the `massE` velocity and MGS density catalogs. Blue and orange contours denote the results using covariances estimated from 400 and 169 jackknife subsamples, respectively, and the inner and outer represent enclose the 68% and 95% credible regions. **Top panel**: Constraints from joint analysis of the velocity and density statistics, $\{\xi^{gg}_{0,2}, \xi^{gu}_1, \xi^{uu}_0\}$. **Bottom left panel**: Constraints from the velocity-only analysis, $\{\xi^{uu}_0\}$. **Bottom right panel**: Constraints from the density-only analysis, $\{\xi^{gg}_{0,2}\}$. Although the posterior distributions exhibit mild shifts as the number of jackknife subsamples varies, the marginalized constraints and the remain stable compared to the statistical uncertainties.

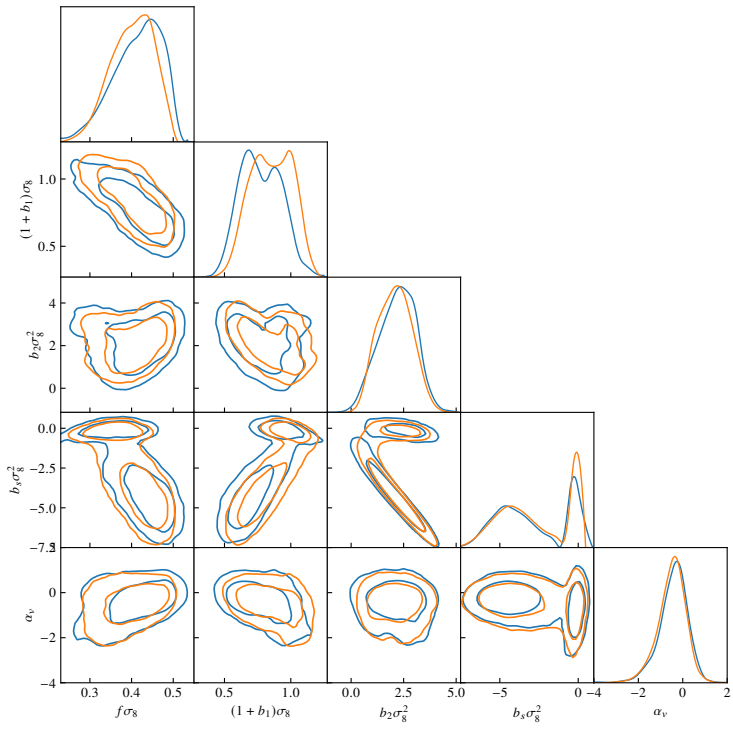


FIG. 33. Posterior distributions of the model parameters from the joint analysis of $\{\xi_{0,2}^{gg}, \xi_1^{gu}, \xi_0^{uu}\}$ measured from the `massE` velocity and MGS density catalogs. Same as Fig. 32, but we restrict the fitting range to $s \in [20, 120]\,\mathrm{Mpc}/h$ instead of $s \in [20, 160]\,\mathrm{Mpc}/h$.

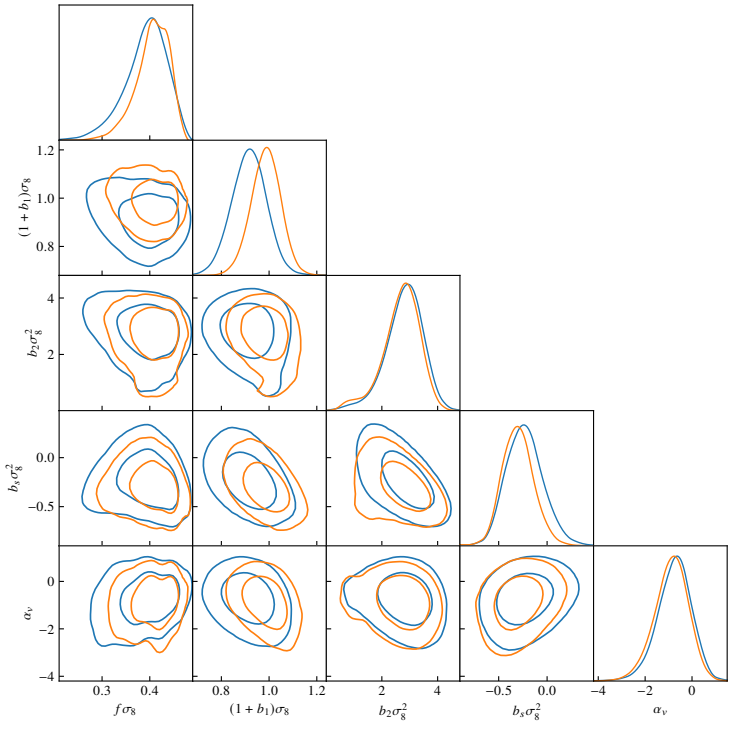


FIG. 34. Posterior distributions of the model parameters from the joint analysis of $\{\xi_{0,2}^{gg}, \xi_1^{gu}, \xi_0^{uu}\}$ measured from the `massE` velocity and MGS density catalogs. Same as Fig. 32, but we apply the additional shuffle-subtraction procedure to the measurements and the corresponding model predictions.

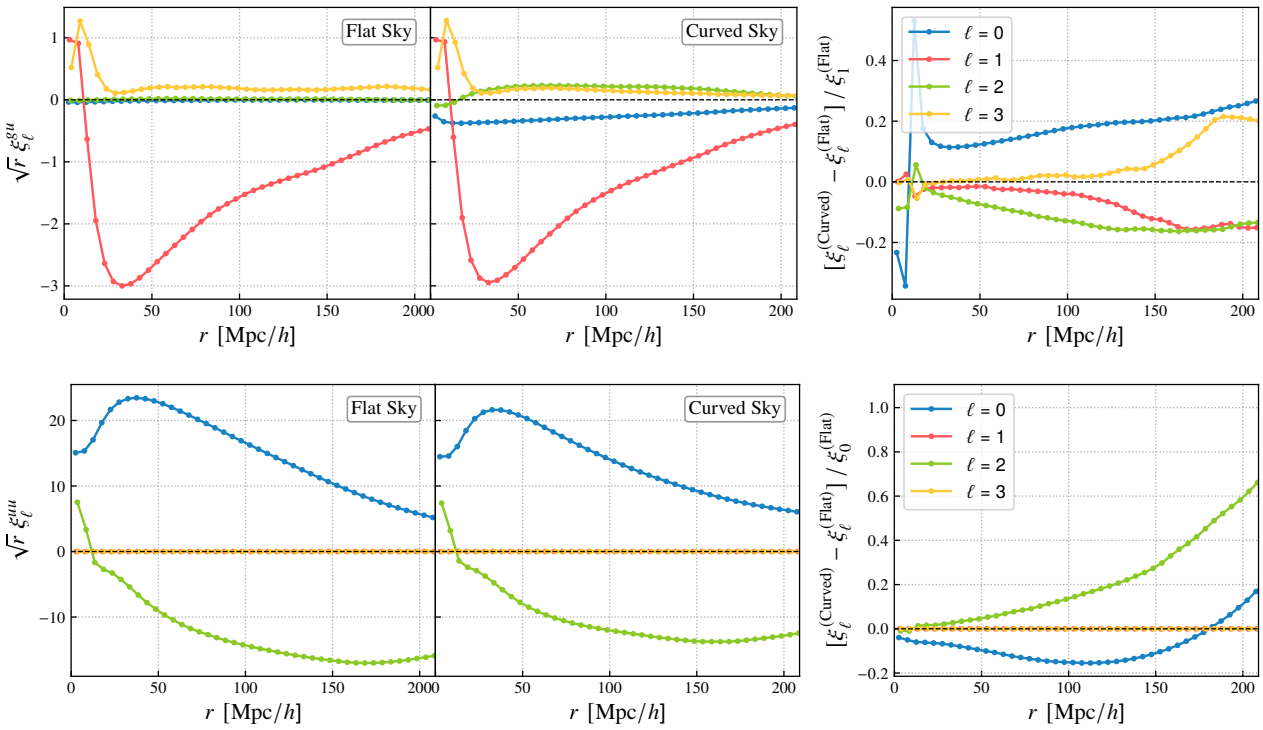


FIG. 35. **Estimation of the wide angle effects on the redshift-space velocity statistics using N-body simulations.** The top panel shows the matter-velocity cross-correlation $\xi_\ell^{gu}(s)$, while the bottom panel shows the velocity auto-correlation $\xi_\ell^{uu}(s)$. The columns from left to right show the redshift-space multiples with flat sky approximation, the redshift-space multiples with curved sky, and their relative difference normalized by the leading non-vanishing multipole.

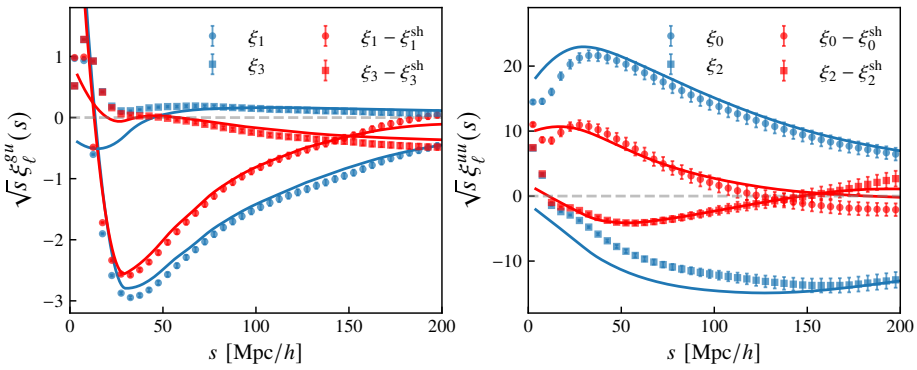


FIG. 36. **Effect of the shuffle-subtraction procedure on the velocity statistics.** We show the velocity statistic multipoles measured from the dark matter mock (points with error bar) and the perturbation theory predictions (solid lines). The left and right panels show the density-velocity cross-correlation ξ_ℓ^{gu} and the velocity auto-correlation ξ_ℓ^{uu}, respectively. Blue and red colors denote results before and after applying the shuffle-subtraction procedure, respectively.